“Identification of Early-Stage Biomarkers in Epithelial Ovarian Cancer Through Integrative Transcriptomic and Co-Expression Network Analysis”

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that this project entitled **"Identification of Early-Stage Biomarkers in Epithelial Ovarian Cancer Through Integrative Transcriptomic and Co-Expression Network Analysis"** is a Bonafide record of work done by KOLLAPUR DHAKSHA PRAGNYA SREE (Reg.No:24MSBINPY0024),MSc..in Bioinformatic at PONDICHERRY UNIVERSITY, Pondicherry, has been carried out project under my direct supervision and guidance. Certified further that to the best of my knowledge, the work reported herein does not form part of any other work or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

I hereby declare that the project entitled " **Identification of Early-Stage Biomarkers in Epithelial Ovarian Cancer Through Integrative Transcriptomic and Co-Expression Network Analysis**" is submitted for completion of Project in MSc.Bioinformatics. This work is entirely an original one done by me at the Department of Bioinformatics, Pondicherry University, Pondicherry. This work has not been submitted elsewhere for any other degree, diploma..

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Place: Puducherry
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KOLLAPUR DHAKSHA PRAGNYA SREE

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

Abstract

Ovarian cancer that develops in the epithelium of the ovaries is classified as the most deadly form of gynecologic cancer primarily due to the late diagnosis and ongoing lack of reliable biomarker discovery for early detection. The objective of the work presented is to identify gene expression signatures at cancer stage and putative biomarkers through an integrated transcriptomic analysis approach. The RNA-seq data from The Cancer Genome Atlas (TCGA) and Genotype-Tissue Expression (GTEx) has been pooled together, and batch effects removed utilizing surrogate variable analysis (SVA). Principal Component Analysis (PCA) was employed to provide an initial assessment of data quality and to examine the clustering of samples within the RNA-seq dataset. The examined differential gene expression analyses utilized DESeq2 software and identified a total of 1,931 upregulated genes at the early stage of disease and 1,999 upregulated genes at the late stage of disease, of which, 1,739 or 90% are common to both disease stages. The high number of genes that are upregulated across stage indicates that there are significant tumor-associated alterations which occur early in cancer development, and remain as the tumor progresses. A total of 23 genes were found to have differential gene expression between the two stages of disease, suggesting that there are a small number of key regulatory genes that drive the progression of Epethelial Ovarian Cancer. Genes RERGL, EN2, ITLN2 and TMEM132B had substantial changes in expression, and identified as potential biomarkers. Weighted Gene Co-expression Network Analysis (WGCNA) enabled the identification of stage-specific modules associated with EOC, with the majority of early stage modules enriched in genes within cell cycle pathways and a majority of late stage modules enriched in metabolic processes including oxidative phosphorylation. Ultimately, the results of this study provide compelling evidence that common molecular mechanisms can be used to explain the biological processes associated with EOC

CHAPTER-1
INTRODUCTION

1. Introduction

1.1 Epithelial Ovarian Cancer

Epithelial ovarian cancer is often called the “silent killer” for a reason-it’s the deadliest gynecological cancer out there. In the early stages, the signs are so subtle that most women have no idea anything’s wrong. By the time something actually feels off, the cancer has usually spread. According to Webb and Jordan (2024), more women die from ovarian cancer than from cervical and endometrial cancers combined. The five-year survival rate is still stuck under 40%. It’s bleak, honestly. Over 75% of women find out they have it only after it’s already reached advanced stages (FIGO stage III or IV). At that point, treatment options shrink, and survival odds drop sharply(Lheureux et al., 2019).

Right now, doctors rely on serum biomarkers like CA-125 to look for ovarian cancer, but these just aren’t cutting it. They’re too vague-lots of benign conditions can send CA-125 up, so it’s anything but specific. It also misses too many early cases. Bastos et al. (2024) and Zhang et al. (2022) both agree: our current screening isn’t good enough to catch ovarian cancer while there’s still time to do something about it. The Risk of Ovarian Malignancy Algorithm (ROMA), which combines CA-125, HE4, and some symptom tracking, works a bit better, but still can’t catch enough cases early. Multi-omics research has found some interesting molecular signatures (Liu et al., 2020), but most of these discoveries aren’t ready for the clinic yet.

Diagnostic performance of biomarkers and combined modalities in ovarian cancer detection.

Markers	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)
CA-125 [88]	80.1	53.6	48.4	83
CA-125 + MI [85]	98.1	80.8	40.9	99.7
ROCA, CA-125 [87]		92	4.6	
ROCA, CA-125 + TVUS [89]		99.9	40	
CA 125 + symptoms [90]	89.3	83.5		
CA-125 + HE4 + symptoms [93]	79	91	83	89

Table-1 **Diagnostic Performance of Biomarkers and Combined Modalities in Ovarian Cancer Detection**(Dochez et al., 2019).

1.2 The Biomarker: From CA125 to Novel Candidates

For now, CA-125 is still the workhorse for ovarian cancer. It actually tracks a molecule called MUC16. But CA-125 isn't great for early disease. (Charkhchi et al. (2020))found that MUC16 doesn't just "mark" the tumor-it helps it spread, especially by sticking to mesothelin and dodging the immune system. The bigger issue? CA-125 just misses too many early cases. Yousefi et al. (2021) showed that half of women with early-stage ovarian cancer have totally normal CA-125 results.

Because of all this, researchers have gone looking for better, more reliable biomarkers. They're turning to big databases like TCGA and GEO, because you need a lot of data to tease out subtle signals. For instance, (Wang et al. (2022)) discovered that PTH2R is much more active in ovarian cancer. Their multi-omics work, which combined gene expression, qPCR, and Western blots, suggests PTH2R actually helps drive tumor growth-not just mark it-which could make it useful for both diagnosis and treatment.

1.3 Integrating Bioinformatics and Machine Learning

Bioinformatics and machine learning are starting to shake things up, too. Algorithms like Random Forest, XGBoost, and CatBoost are giving us cancer models with more punch. Now, thanks to tools like SHAP, researchers can see which genes actually make a difference. (Chekini et al. (2025))used SHAP to home in on genes like SGO1, VTA1, and RBM5-AS1-turns out, these really steer how ovarian cancer develops.

At the same time, non-invasive methods are becoming more popular. Liquid biopsies—using blood or other fluids—are in the spotlight. Recent studies (Hulstaert et al., 2021) show that bits of DNA, cell-free RNA, and even lncRNAs in the blood could help with earlier detection and tracking the disease.

That's where this study comes in. By merging big data from TCGA and GTEx, the goal is to pin down new gene signatures that reliably spot ovarian cancer early. That's the only real way to turn things around for patients.

1.4 Research Gap

But in Reality, for all the technology and medical word revolving around new markers, early detection is still a question. The old amarkers-CA-125, HE4-just aren't sensitive enough for early disease, and false alarms happen way too often. While combination strategies like ROMA look better on paper, they still don't give us the effective screening tool we desperately need.

High-throughput sequencing and multi-omics studies have uncovered a high of possible new markers. Still, most are linked to late-stage cancer, which doesn't help save lives. Plus, the studies themselves aren't easy to compare or repeat-everyone uses different datasets, methods, and sample sizes. And even though there's tons of transcriptomic data in databases like TCGA and GTEx, researchers haven't fully mined them for early-stage signals. Most just look at whether a gene is up or down-not if groups of genes are moving together or what that means for how cancer works.

Machine learning is helping, but you rarely see it paired with deeper network-based approaches like WGCNA. Most markers still aren't validated enough to reliably distinguish early from late-stage cancer, which really matters for patient survival.

So here's the real gap: we're missing solid, biologically meaningful biomarkers that can spot ovarian cancer early-before it's too late. To get there, we need a new approach. That means analyzing gene expression, tapping into big data, using network-based strategies, and pulling it all together to boost early detection and survival. That's the direction this study is headed.Building upon these advancements, the current study addresses the urgent clinical need for reliable early-stage biomarkers, as many existing indicators are only dominant in late-stage malignancy. By integrating the TCGA and GTEx datasets, this research aims to identify novel biomarkers specifically tailored for the early detection of ovarian cancer

CHAPTER 2

OBJECTIVES

2. Objectives of the Study

The present study aims to identify potential biomarkers for early detection of epithelial ovarian cancer using integrative bioinformatics approaches. The specific objectives are:

1. To integrate and preprocess transcriptomic datasets from TCGA and GTEx to obtain a comprehensive and comparable gene expression profile of ovarian cancer and normal tissues.
2. To perform differential gene expression analysis (DEGs) to identify significantly upregulated and downregulated genes between ovarian cancer and normal samples, including stage-specific comparisons (early vs. late stages).
3. To construct gene co-expression networks using Weighted Gene Co-expression Network Analysis (WGCNA) and identify key gene modules associated with clinical traits, particularly early-stage ovarian cancer.
4. To perform functional enrichment analysis (GO and KEGG pathway analysis) to understand the biological processes and pathways associated with the identified genes and modules.
5. To explore the potential of identified genes as early-stage diagnostic markers, distinguishing them from genes predominantly expressed in late-stage disease.

CHAPTER-3

METHODOLOGY

3. Methodology

TCGA FOR TUMOR AND GTEX FOR NORMAL
↓
Create metadata
↓
DESeqDataSetFromMatrix
↓
EstimateSizeFactors
↓
Run SVA(hidden batch detection)
↓
Add SVs to design
↓
DESeq()
↓
Extract results
↓
Dysregulated Early vs Normal
Dysregualted Late vs Normal
Dysregulated Early vs Late
↓
Filter significant genes
↓
Extract top up/down genes
↓
Volcano plot
↓
KEGG / GO enrichment
↓
WGCNA

Fig-1 Work Flow of the Project

3.1. Data Acquisition and Pre-processing

Tool: TCGAbiolinks *(Colaprico et al., 2016)* and recount3 ((Wilks et al., 2021)

TCGAbiolinks is an R/Bioconductor package designed to facilitate the retrieval and analysis of data from the GDC portal. recount3 provides access to uniformly processed RNA-seq data from various projects including GTEx.

Tumor samples for the TCGA-OV project were downloaded using GDCdownload, and clinical metadata was integrated via GDCprepare. Normal tissue samples from the ovary were retrieved from the GTEx project using recount3.The raw data included 19,962 protein-coding genes across 280 TCGA samples. After tissue filtering and data cleaning, the final count matrix contained 17,230 genes for TCGA. The GTEx dataset initially contained 22,321 genes across 195 samples.

```
cat("Genes after protein coding filter:", nrow(tcga_counts), "\n")Genes after protein codi
ng filter: 19962
```

```
Starting to add information to samples
 => Add clinical information to samples
 => Adding TCGA molecular information from marker papers
 => Information will have prefix 'paper_'
Available assays in SummarizedExperiment :
  => unstranded
  => stranded_first
  => stranded_second
  => tpm_unstrand
  => fpkm_unstrand
  => fpkm_uq_unstrand
```

Fig-2 R code

GTEX	**TCGA**
Unfiltered: 63,856	**Unfiltered**: 60,676
Only protein coding genes: 22,321	**Only protein coding genes**: 19,962
	Not reported removed: Total 276 samples with stages
Low count filtering: (0 > 10 should be present)	**Low count filtering**: (0 > 10 should be present)
Total: 17,147	**Total**: 17,707

Table-2 Dataset filtering

Common genes matched total of 15,873 remaining were removed as they are only present in one of them and after merging samples 471

Those need to be extracted according to stage and to verify either in which samples tcga or getx normal they are not present

Those were removed to ensure crossplatform comparability only genes common to both TCGA and GTex reference annotation were retained. This is important to prevent technical artifacts during batch correction

3.2 .PCA (Principal Component Analysis)

Tool: PCA (Jolliffe & Cadima, 2016),R

Principal Component Analysis (PCA) is a widely used dimensionality reduction technique that transforms high-dimensional gene expression data into a smaller set of orthogonal components (principal components) that capture the maximum variance in the dataset. PCA helps in visualizing sample clustering patterns, identifying outliers, and assessing the presence of batch effects in transcriptomic data.

Method:Principal Component Analysis (PCA) was performed to evaluate the overall variance and sample distribution in the integrated RNA-seq dataset. PCA was used both before and after batch effect correction using SVA to assess the presence of technical variation and the effectiveness of normalization. The analysis enabled visualization of clustering patterns between tumor and normal samples, confirming improved data consistency after correction while preserving biologically relevant differences.

3.3. Integration and Confounding Effect

Method: Batch Effect Assessment and SVA (Leek et al., 2012)

Batch effects refer to unwanted technical variations introduced during data generation, such as differences in sequencing platforms or laboratory conditions.In this study, the dataset variable (GTEx vs. TCGA) was perfectly confounded with the biological condition (Normal vs. Tumor), as all normal samples originated from GTEx and all tumor samples from TCGA

Surrogate Variable Analysis (SVA): To account for hidden sources of variation without modifying raw counts, SVA was employed. Unlike ComBat, which was avoided for differential expression to prevent removing genuine biological signals, SVA identifies surrogate variables that capture variation not explained by the primary model. A subset of 10 surrogate variables was included in the DESeq2 design formula to balance noise reduction with the preservation of biological signals.(Leek et al., 2012)

First they are added to the metadata:

```
metadata$SV1 <- svseq$sv[,1]

metadata$SV2 <- svseq$sv[,2]
design = ~ SV1 + SV2 +…..+stage group
```

This means the model now becomes:

Gene Expression ~ Hidden Variation + Biological Condition

So DESeq2 estimates differential expression after accounting for the hidden variation.

3.4. Differential Gene Expression Analysis

Tool: DESeq2（Love et al., 2014)

DESeq2 is a standard R package used to perform differential expression analysis using a model based on the negative binomial distribution.

A DESeqDataSet was constructed from the combined count matrix of 15,873 common genes. Comparisons were performed across three groups: Early vs. Normal, Late vs. Normal, and Early vs. Late. Genes were identified as differentially expressed (DEGs) if they met a threshold of Padj<0.05 and log2FC ≥2.

The analysis identified 3,949 DEGs for Early vs. Normal and 3,911 DEGs for Late vs. Normal. A smaller set of 116 DEGs was found when comparing Early vs. Late stages.

```
> dds <- DESeq(dds)
 estimating size factors
 estimating dispersions
 gene-wise dispersion estimates
 mean-dispersion relationship
 final dispersion estimates
 fitting model and testing
 -- replacing outliers and refitting for 906 genes
 -- DESeq argument 'minReplicatesForReplace' = 7
 -- original counts are preserved in counts(dds)
 estimating dispersions
 fitting model and testing
> dim(dds)
[1] 15873   471
```

Fig-3 Deg R code

Condition	Deg criteria	Up_regulated	Down_regulated
early_vs_normal	3949 genes	1931	2018
late_vs_normal	3911 genes	1999	1912
Early_vs_late	116 genes	23	93

Table -3 Deg analysis table

3.5. Functional Enrichment and Pathway Analysis

Tool: Enrichr *(Kuleshov et al., 2016)* and GEPIA (Tang et al., 2017)

Functional enrichment analysis identifies biological pathways and gene ontology (GO) terms overrepresented in a gene set. Enrichr is a comprehensive gene set enrichment analysis tool, and GEPIA is used for validating gene expression levels.

Differential Expression Analysis (DEA) was performed to identify genes associated with disease progression across three stages: Early vs. Normal, Late vs. Normal, and Early vs. Late. The top 300 significant DEGs from each stage were exported for KEGG and GO (Biological Process, Cellular Component, Molecular Function) analysis. For the 23 upregulated genes in the Early vs. Late comparison, expression levels were cross-checked with GEPIA to confirm they showed expression in normal tissue but not in tumor samples.

3.6 GO and KEGG Pathway Enrichment

Functional annotation and pathway enrichment were conducted using Enrichr, a comprehensive gene set enrichment analysis web server. The gene lists were mapped against two primary databases:

Gene Ontology (GO): Genes were categorized into three domains: Biological Process (BP), Cellular Component (CC), and Molecular Function (MF). This provided a multi-layered view of what the genes do, where they are located, and their biochemical activity.

KEGG (Kyoto Encyclopedia of Genes and Genomes): This was used to identify higher-order systemic functions and molecular interaction networks (pathways) associated with the gene sets.

3.7 Protein-Protein Interaction (PPI) Network

Tool: STRING Database (Szklarczyk et al., 2021) **and Cytoscape** (Shannon et al., 2003)

The STRING database provides known and predicted protein-protein interactions. Cytoscape is an open-source platform for visualizing and analyzing complex networks.

A PPI network was constructed for the top 300 DEGs using the STRING database (v12.0). The DEGs were mapped with a "High" confidence score requirement of 0.700. The resulting interactome was imported into Cytoscape (v3.10) and laid out using the Prefuse Force Directed Layout to visually separate functional clusters.

3.7 Hub Gene Identification

Tool: cytoHubba (MCC Algorithm) (Chin et al., 2014)

CytoHubba is a plugin for Cytoscape used to calculate the importance of nodes in a network. The Maximal Clique Centrality (MCC) algorithm is a local ranking method that identifies essential biological "hubs".Within cytoHubba, the MCC algorithm was selected to rank the genes. The top 25 genes were extracted and visualized in a circular layout, with a color gradient (Red to Yellow) representing the MCC score intensity.

3.8 Weighted Gene Co-expression Network Analysis (WGCNA)

Data Preprocessing (DESeq2 - VST)
↓
Expression Matrix (Samples × Genes)
↓
Filtering Low Variance Genes
↓
Quality Check (goodSamplesGenes)
↓
Soft Threshold Selection (pickSoftThreshold)
↓
Selected Power $\beta = 6$
↓
Topological Overlap Matrix (TOM)
↓
Hierarchical Clustering of Genes
↓
Module Detection (Color-coded modules)
↓
Module Eigengene Calculation (MEs)
↓
Module–Trait Correlation
↓ ↓
Early vs Normal Late vs Normal
↓ ↓
Significant Modules Significant Modules
↓ ↓
Gene Extraction Gene Extraction
↓
Differential Expression Analysis (DESeq2)
↓
Module–DEG Overlap Analysis
↓
Top Genes Selection (logFC + significance)
↓
Heatmap Visualization
↓
Hub Gene Identification
↓
Pathway analysis

Fig-4 WGCNA detailed Workflow

Tool: WGCNA Package (Langfelder & Horvath, 2008)

WGCNA is a systems biology method used to identify clusters (modules) of highly correlated genes and relate them to external traits.To identify stage-specific gene clusters in ovarian cancer, we employed the WGCNA package (v1.72) in R (Langfelder & Horvath, 2008).

1. **Data Preprocessing:** RNA-seq count data were normalized using the Variance Stabilizing Transformation (VST) via the DESeq2 package to account for heteroscedasticity (Love et al., 2014). To reduce noise, the bottom 25% of genes with the lowest variance were filtered out. The goodSamplesGenes function was used to ensure data integrity.

	Power	SFT.R.sq	slope	truncated.R.sq	mean.k.	median.k.	max.k.
1	1	0.364	0.234	0.328	4730.0	4.91e+03	8370
2	2	0.844	-0.474	0.833	2170.0	1.95e+03	5610
3	3	0.914	-0.705	0.933	1210.0	8.61e+02	4150
4	4	0.940	-0.831	0.971	749.0	4.03e+02	3240
5	5	0.949	-0.917	0.982	499.0	1.96e+02	2630
6	6	0.958	-0.977	0.991	350.0	9.91e+01	2190
7	7	0.965	-1.020	0.995	255.0	5.13e+01	1850
8	8	0.970	-1.060	0.995	191.0	2.73e+01	1600
9	9	0.974	-1.090	0.995	147.0	1.48e+01	1400
10	10	0.977	-1.100	0.994	116.0	8.12e+00	1230
11	12	0.982	-1.130	0.993	75.5	2.61e+00	987
12	14	0.986	-1.140	0.993	52.0	9.16e-01	814
13	16	0.988	-1.150	0.991	37.3	3.32e-01	686
14	18	0.991	-1.150	0.992	27.8	1.31e-01	589
15	20	0.992	-1.140	0.992	21.3	5.13e-02	513

Table-4 Soft power pick

2. **Network Construction a** weighted adjacency matrix was calculated based on the Pearson correlation between gene expression profiles. To satisfy the scale-free topology criterion, we evaluated soft-thresholding powers=6 from 1 to 20 using the pickSoftThreshold function. A power of beta = 6was selected as it reached a scale-free fit index (R^2) of **0.958**.

3. **Module Identification** the adjacency matrix was transformed into a Topological Overlap Matrix (TOM) to calculate gene interconnectedness. And modules were defined via the Dynamic Tree Cut method with a minimum module size of 30 genes (Langfelder et al., 2008).

4. **Trait Correlation and Integration** module Eigengenes (MEs) were calculated as the first principal component of each module. These were correlated with clinical stages (Normal vs. Early and Normal vs. Late) using Pearson correlation. Significant modules were defined by $r > 0.5$ and $p < 0.05$. Finally, modules were intersected with differentially expressed genes (DEGs) identified by DESeq2 log_2FC > 2, padj < 0.05) to pinpoint biologically active networks.

3.9 Downstream Analysis:

Tools: BiomaRt package, R

Once the stage-specific modules were identified, subsequent analyses were carried out to extract genes with biological relevance and evaluate their functional importance. The significant modules linked to clinical traits (Early vs Normal and Late vs Normal) were initially divided into groups based on positive and negative correlations according to the direction of module–trait correlations. Gene lists for each selected module were obtained by using the assigned module colors. To find functionally active genes within these modules, an overlap analysis was conducted between the genes in the modules and the differentially expressed genes (DEGs) identified from DESeq2 analysis ($|\log_2 FoldChange| > 2$ and adjusted p-value < 0.05). Only those genes found to overlap were retained for further examination, as these represent genes that are both co-expressed and significantly dysregulated.

For each module, the overlapping gene sets were extracted for subsequent functional enrichment analysis. Gene annotation was achieved by translating ENSEMBL gene identifiers to gene symbols with the help of the biomaRt package (Ensembl database). Functional enrichment analysis utilized KEGG pathway analysis to pinpoint biological pathways that were significantly enriched within each gene set specific to the modules. To illustrate expression patterns, heatmaps were created using the expression profiles of DEGs that overlapped from chosen modules. Before visualization, expression data were standardized (Z-score normalization) across samples, and genes with missing data or no variance were eliminated. Hierarchical clustering was employed for both genes and samples to evaluate grouping tendencies. Annotations of samples were included to differentiate between clinical stages.

Furthermore, for every module, the quantity of overlapping genes with differentially expressed genes (DEGs) was assessed to determine the significance of each module. Modules that exhibited a greater overlap with DEGs were given priority for additional interpretation and visualization. Distinct analyses were performed for Early versus Normal and Late versus Normal comparisons to uncover transcriptional signatures specific to each stage.

-

CHAPTER 4
RESULTS AND DISCUSSIONS

RESULTS AND DISCUSSIONS

4.1 Differential Gene Expression Analysis

The differential expression analysis was performed using the DESeq2 package in R to identify significant biomarkers across different stages of ovarian cancer. The analysis utilized a stringent threshold of an adjusted p-value (P_adj) < 0.05 and a log_2 Fold Change (log_2FC) to ensure high-confidence results.

X-axis: log_2 Fold Change log_2FC

This axis measures the magnitude and direction of the change in gene expression

Y-axis: Negative log_10 Adjusted P-value

This axis represents the statistical significance of the change in gene expression

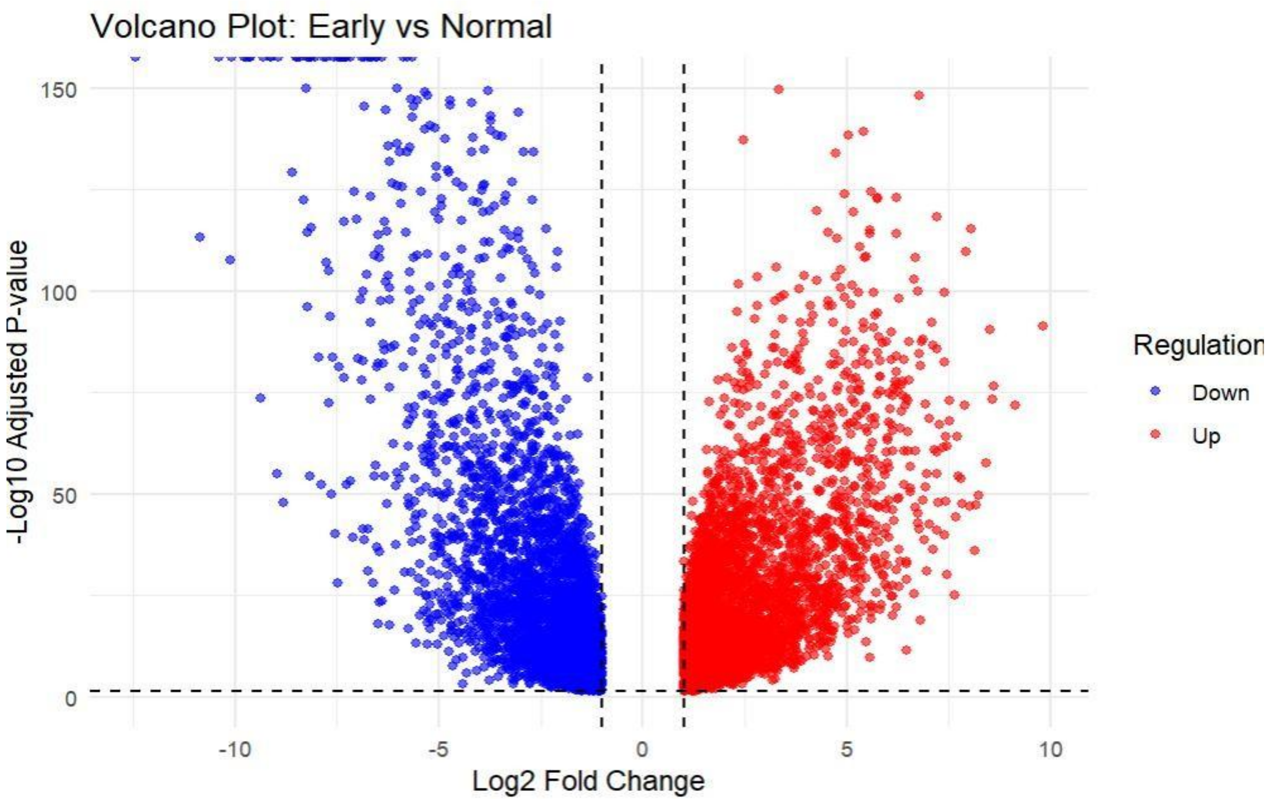
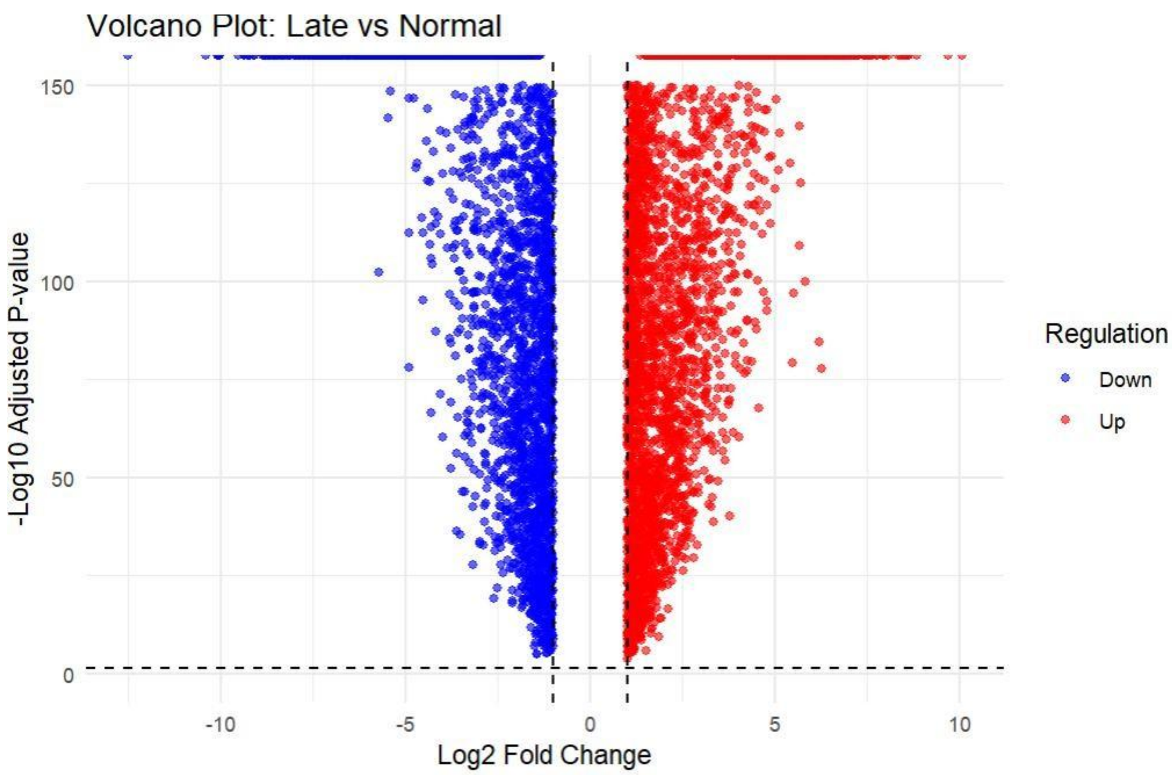


Fig- 5 Early Stage vs. Normal up regulated(Red)and downregulated(Blue) gene expression

Total DEGs: A total of 3,949 differentially expressed genes were identified.Regulation Pattern: Out of these, 1,931 genes were significantly upregulated, and 2,018 genes were downregulated. The plot shows a robust distribution of significant genes, with many exceeding a -log_10 Adjusted P-value of 150, indicating highly significant biological variation during the early transition to malignancy.

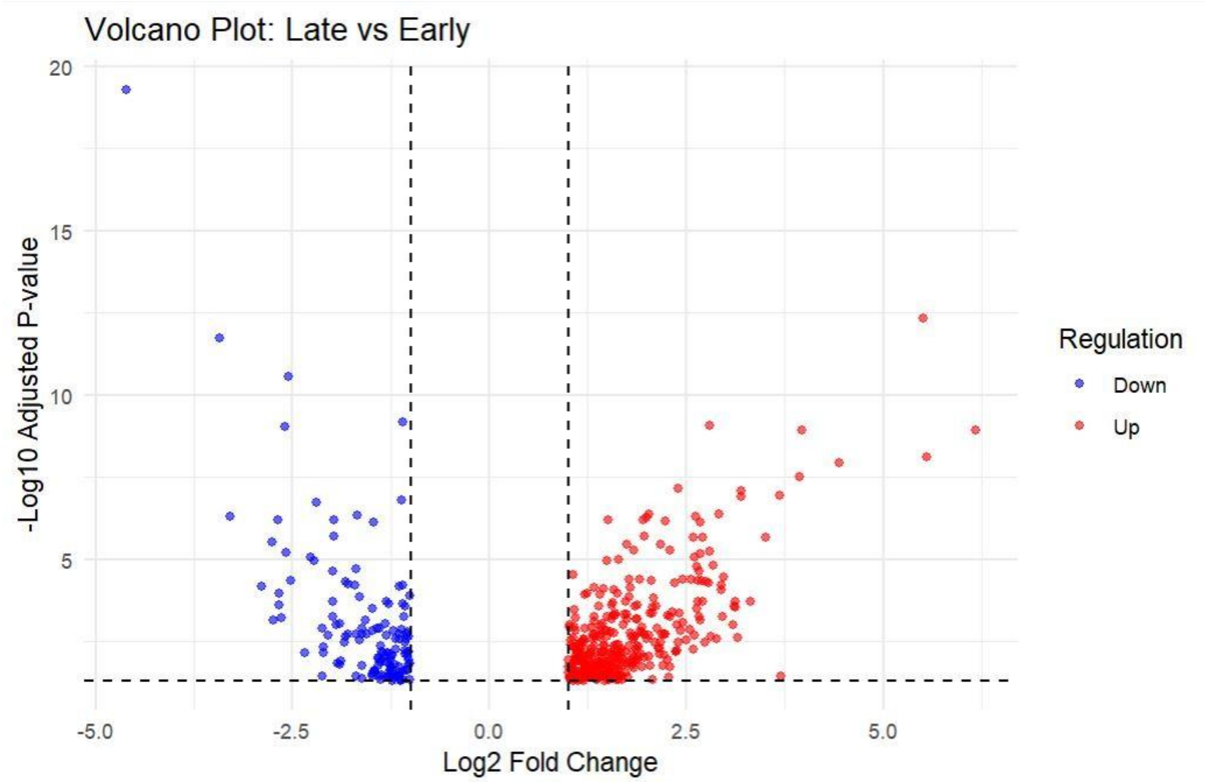


Late vs Normal Deg Expression:

Figure 6. Late Stage vs. Normal up regulated(Red)and downregulated(Blue) gene expression

Total DEGs: A total of **3,911** differentially expressed genes were identified.This stage showed **1,999** upregulated genes and **1,912** downregulated genes. Similar to the early stage, the late stage shows extensive dysregulation. The plot shows a robust distribution of significant genes, with many exceeding a -log_10 Adjusted P-value of 150, indicating highly significant biological variation in late stage

Early Stage vs. Late Stage



.

Fig 7. Early Stage vs.Late up regulated(Red)and downregulated(Blue) gene expression

Total DEGs: This comparison yielded a significantly smaller set of **116** genesOnly **23** genes were significantly upregulated in late stages compared to early stages, while **93** genes were downregulated.

The relatively low number of DEGs (compared to the normal tissue comparisons) is seen while the progression from early to late stage involves more subtle, specific shifts in gene expression.

Condition	Deg criteria	Up_regulated	Down_regulated
early_vs_normal	3949 genes	1931	2018
late_vs_normal	3911 genes	1999	1912
Early_vs_late	116 genes	23	93

4.2 Venn Diagram

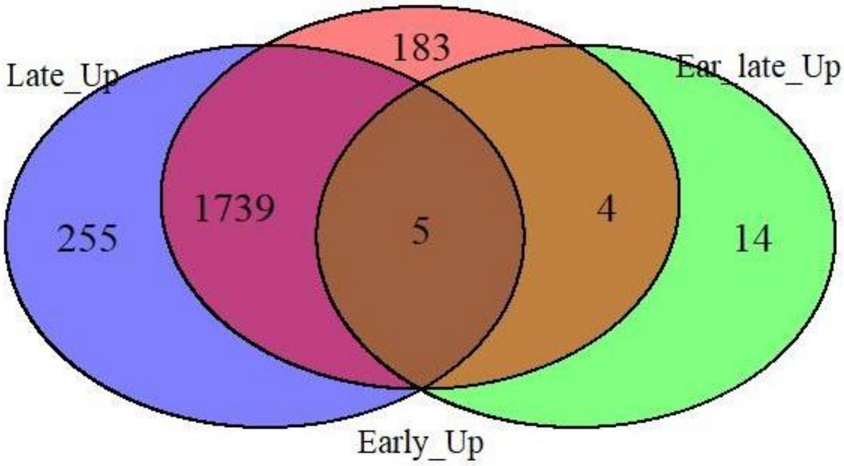


Fig 8 Venn Diagram of Upregulated Genes

A) Early Vs Normal

1. The Early vs Normal Specific Markers (183 Genes)

The 183 genes in the non-overlapping part of the Early_Up circle are only found in the early-stage tumor and not in healthy tissue. From a research standpoint, these represent your most significant early detection biomarkers. They are probably part of the "initiation" phase of the cancer because they are only upregulated in the early phase and not very much in the late-stage comparison.

2. The Core Tumor Signature (1,739 Genes)

The Late_Up group shares 1,739 genes with the Early_Up set. This big overlap is what makes up the "Core Cancer Signature." These genes are always turned on, from the time the tumor is first found until it is very advanced. In biological terms, these genes usually control basic features of cancer, such as speeding up the cell cycle, fixing DNA, and changing how the body uses energy (like the Warburg effect). They are not useful for telling the difference between stages, but they are great targets for broad-spectrum therapeutic intervention.

3. The Progression Drivers (4 + 5 Genes)

The parts of the Early_Up circle that overlap with the "Early vs. Late" comparison are small but important.

The Overlap of 4: These 4 genes are more active in early stages compared to normal. They also show a significant change when comparing the early stage directly to the late stage. These likely represent genes that are "switched on" early but are adjusted significantly as the tumor develops.

The "All-Stage" Center (5 Genes): These 5 genes are the strongest markers in your study. They are more active in early versus normal, late versus normal, and they show significant differences between early and late. These are your Progression Biomarkers. They are present from the beginning, but their expression level changes significantly as the disease progresses, making them ideal for tracking stage advancement.Early_vs_Normal = Early only (183)+ Early ∩ Late (1739)+

Early ∩ Early_vs_Late (4)+ All three (5)

= 183 + 1739 + 4 + 5

= 1931

B) Late Vs Normal

The **Late vs Normal** group (Total = 1,999 genes) represents the molecular state of the tumor in its advanced phase

1. The Late-Specific "Aggression" Markers (255 Genes)

The 255 genes unique to the Late_Up circle are significantly upregulated only in the late stage compared to normal tissue. These genes are often linked to metastasis, invasion, and therapy resistance. While the early-stage genes focus on initiating the tumor, these late-specific genes usually control:

Immune Evasion: They develop advanced ways to evade the body's natural killer cells. Their presence serves as excellent prognostic markers, indicating that the disease has progressed beyond its original site.

2. The Persistent Core (1,739 Genes)

Like the early stage, most of the Late_Up set consists of 1,739 genes shared with the Early_Up group. This shows that the "identity" of the cancer remains stable. Even in the late stage, the tumor still relies on the same fundamental issues, such as rapid cell division and metabolic changes, that were evident at the beginning. These genes serve as the housekeeping components of the tumor; they are not useful for staging but are necessary for the tumor's survival.

3. The Stage-Transition Markers (0 + 5 Genes)

Interestingly, the overlap between Late_Up and the "Early vs. Late" comparison is very specific:

The "All-Stage" Center (5 Genes): These 5 genes are high-value indicators of progression. They are upregulated in both stages compared to normal, but they also show a significant increase in expression when comparing Early to Late directly. This suggests that their expression levels likely correlate with tumor burden; the higher the expression of these 5 genes, the more advanced the cancer.

Late = Late only (255)

+ Early ∩ Late (1,739) + All three (5)

= 255 + 1,739 + 5

= 1,999

C) Early vs Late

The Early vs Late group (shown by the green circle) is the smallest but most statistically significant set for understanding disease progression.

1. The Progression-Specific Drivers (14 Genes)

The 14 genes unique to the Ear_late_Up circle are particularly interesting. These genes are significantly upregulated when comparing the Early stage directly to the Late stage. However, they did not meet the significance threshold when comparing Early vs. Normal or Late vs. Normal individually.

Biological Role: These genes often represent "fine-tuning" genes. They might not be the main drivers of cancer, but they serve as accelerants that help the tumor adapt, evolve, and transition between stages. In your research, these genes could be seen as specific genetic "switches" for stage progression.

2. The Early-to-Late Transition Markers (4 Genes)

The 4 genes shared between Early_Up and Ear_late_Up (but missing in Late_Up) are biologically unique.

Biological Role: These genes are upregulated in the early stage compared to normal tissue and show a significant change when moving to the late stage. However, their failure to be significantly upregulated in the Late vs. Normal comparison suggests that these genes act as "Early-Peak" markers. They likely play a crucial role in the initial growth phase but may be downregulated or lose importance as the tumor becomes more established and advanced.

3. The "Gold Standard" Biomarkers (5 Genes) The 5 genes at the center overlap all three categories. **Biological Role:** These are your "Master Regulators." They are upregulated in the early stage, remain upregulated in the late stage, and show a statistically significant increase between those two stages. For a bioinformatics project, these 5 genes serve as the "Gold Standard" for biomarker discovery since their expression levels can likely predict not only the presence of cancer but also the specific stage the patient is in.

Early_vs_Late = Early_vs_Late only (14) + Early ∩ Early_vs_Late (4)

+ Late ∩ Early_vs_Late (4) + All three (5)

= 14 + 4 + 5

= 23 After getting degs from each stage enrichment analysis has been done

4.3 Protein-Protein Interaction Network

After identifying desregulkated genes from each stage now further network analysis has been done

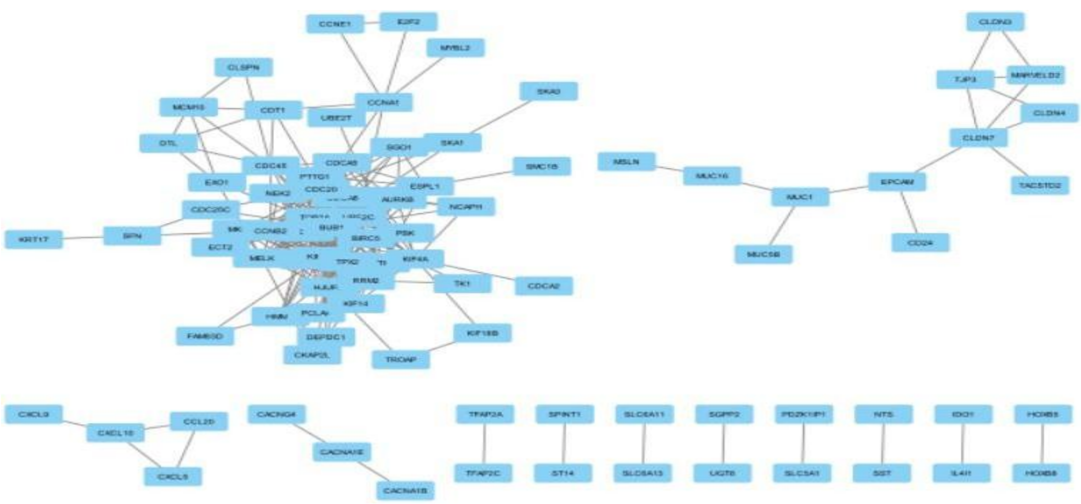


Fig 9:**The global Protein-Protein Interaction (PPI) network of the top 300 DEGs of Early vs Normal.**

Nodes represent proteins, and edges represent functional and physical interactions. The dense central cluster highlights the coordinated activity of genes involved in early-stage disease progression

Early vs Normaltop 25 hub genes

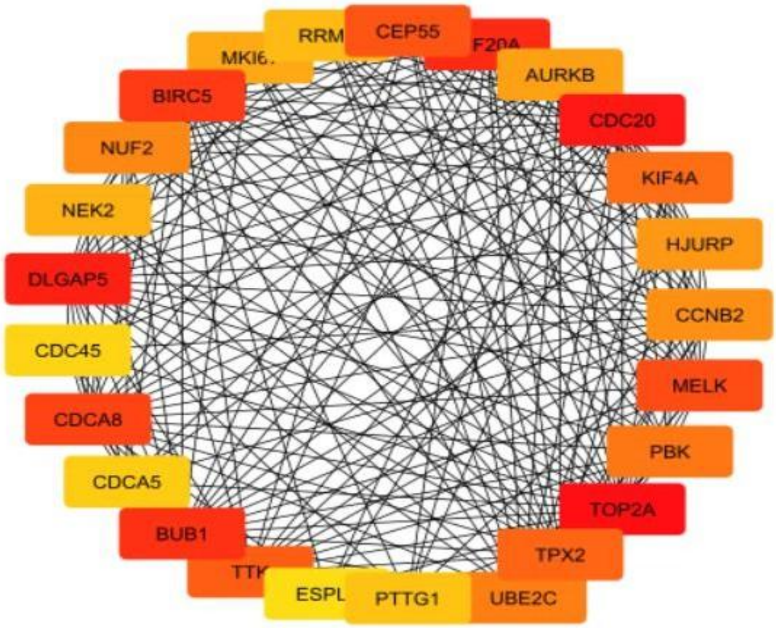


Fig 10: Identification of the top 25 hub genes of Early vs Normal using the cytoHubba MCC algorithm.
The color gradient from red to yellow represents the ranking of essentiality, with red nodes indicating the highest degree of connectivity and biological significance within the network**.**

Network Topology and Hub Gene Identification

The global PPI network was imported into Cytoscape (v3.x) for topological analysis. Figure 9 shows the network has a tightly connected "hairball" structure. This suggests a high level of functional synergy among the upregulated genes. To find the most important regulatory nodes in this dense network, we used the cytoHubba plugin. The Maximal Clique Centrality (MCC) algorithm is known for its accuracy in identifying essential proteins. Using this method, we extracted the top 25 hub genes.

Analysis of Hub Gene Significance

The results of the MCC ranking are shown in Figure 10. In this visualization, node color acts as a heat map for biological importance. Nodes in dark red, like CEP55, BIRC5, and CDCA8, TOP2A, have the highest MCC scores. These represent the key "bottleneck" proteins that keep the network stable. Nodes that shift to orange and yellow, such as PTTG1 and ESPL1, remain important hubs but play a less central role compared to the main red nodes.

Biological Interpretation

From Figure 2, the identified hub genes predominantly belong to the KIF, CDCA, and BUB families. They mainly regulate the mitotic cell cycle, spindle assembly, and chromosome segregation. The fact that these genes are the most central nodes when comparing "early vs. normal" stages suggests that cell cycle

dysregulation is a key event in disease onset. These 25 hub genes, especially the high-scoring red nodes, are prime candidates for early-stage biomarkers. Their central position in the PPI network indicates that their expression levels could act as a "molecular switch." This switch could signal the change from healthy tissue to a pathological state, offering important opportunities for early diagnosis and targeted treatment.1. Top Hub Genes in Late-Stage Pathology

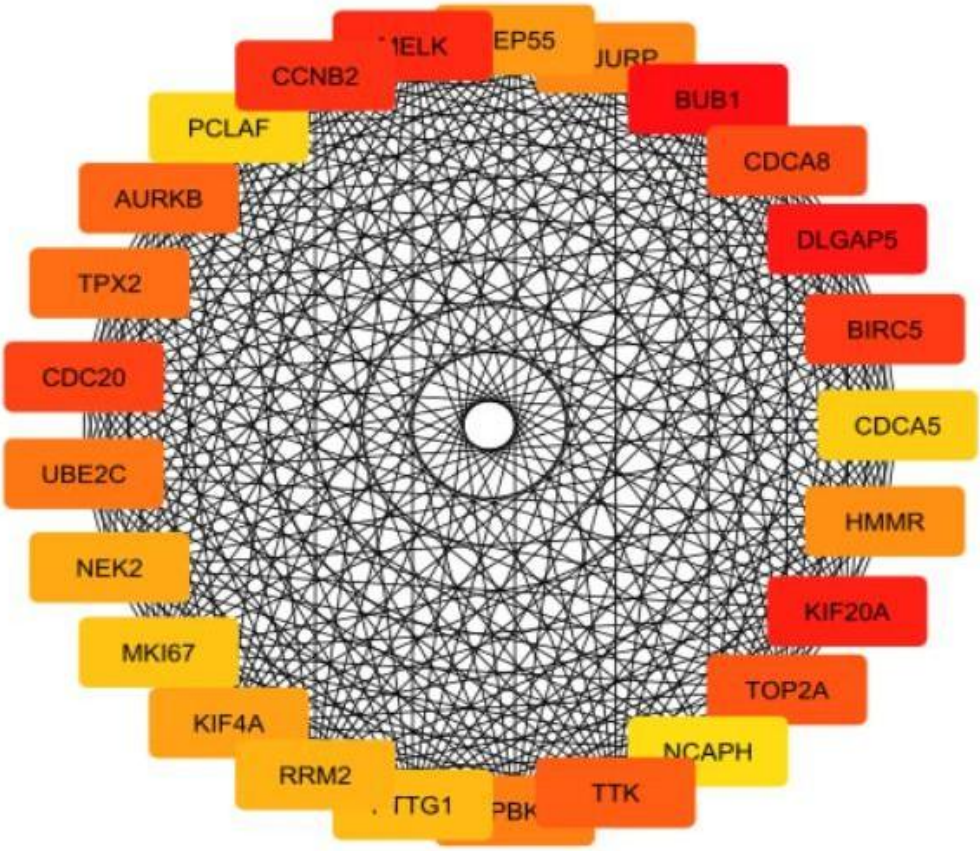


Fig 11: Analysis of late-stage hub genes. Identification of the top 25 hub genes using the MCC algorithm. The shift toward a deeper red color in nodes like *CCNB2* and *BUB1* highlights the intensified reliance of advanced-stage pathology on deregulated mitotic pathways and chromosomal instability.

1. Hub Genes in Late-Stage Pathology

Using the cytoHubba plugin of Cytoscape, the Maximal Clique Centrality (MCC) algorithm was applied to the late-stage interactome. From these calculations, the top 25 hub genes for the late stage interactome are seen in Figure 3 Late-Stage Hub Circles. The network topology in late stage analysis displays more dense high-scoring (Red) nodes than in the early stage analysis. Core hubs include:

Primary Hubs (Dark Red): CCNB2, BUB1, DLGAP5, KIF20A, CDC20 and MELK.

Secondary Hubs (Orange/Yellow): AURKB, TPX2, CEP55, TOP2A, and CDCA8.

2 **Comparative Biological Importance**

MELK and TOP2A were found to be top hub genes present in both the early and late stage networks, and were also known drivers of treatment resistance and DNA replication stress, two parameters typical in the late stage of cancer. The color intensities (Red) of the nodes representing these genes in the late stage network captured their importance in maintaining the network during the samples' progression.

Early vs Late hub genes

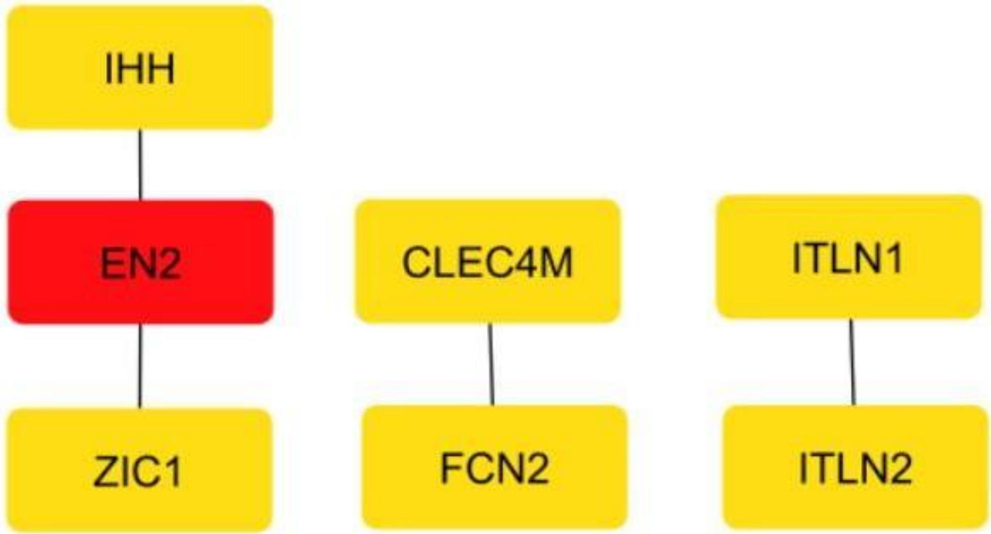


Figure 12: Hub gene analysis of the transition from Early to Late stage disease. Topological analysis using the MCC algorithm highlights *EN2* as the central regulatory "switch" between stages.

1. Core Progression Hubs

As shown in Figure **11: Early vs Normal Hubs]**, the analysis identified a highly specific set of 7 key genes that are differentially regulated as the disease advances. This small, high-confidence cluster represents the "molecular signature" of progression.

The Primary Hub (Red): EN2 (Engrailed Homeobox 2)

EN2 is identified as the highest-ranked hub (MCC score peak) in the progression analysis. *EN2* is a transcription factor typically involved in embryonic development but is frequently "reawakened" in aggressive cancers. Its central position suggests it acts as a master regulator, turning on the pathways required for the late-stage phenotype McGrath, M., et al.[ ] (2013).

○

2. Biological Transition: From Proliferation to Invasion

The transition from the 25 general hubs (seen in earlier) to these specific 7 genes represents a narrowing of focus. While the earlier results were dominated by cell cycle genes, this "Early vs Late" analysis introduces developmental transcription factors (EN2, ZIC1) and immune-modulators (CLEC4M, FCN2).Most of the early vs Late stage genes are showing independebt regulation and importantly three of the genes having high logFC>2 are not invoved in any pathways(ITLN2,TMEM132B,RERGL)

Presence of EN2 along with the other three genes in the early stage is useful for predicting the progression of ovarian cancer.

KEGG pathway analysis of the top 300 upregulated genes revealed enrichment of epithelial differentiation and immune-related pathways.

Cornified envelope formation was significantly enriched in both early and late stages, with higher enrichment in late-stage samples, indicating progressive structural and epithelial alterations. Cell adhesion pathways were prominent in early-stage samples, reflecting initial disruption of tissue integrity.

4.4 KEGG_EARLY_VS_NORMAL

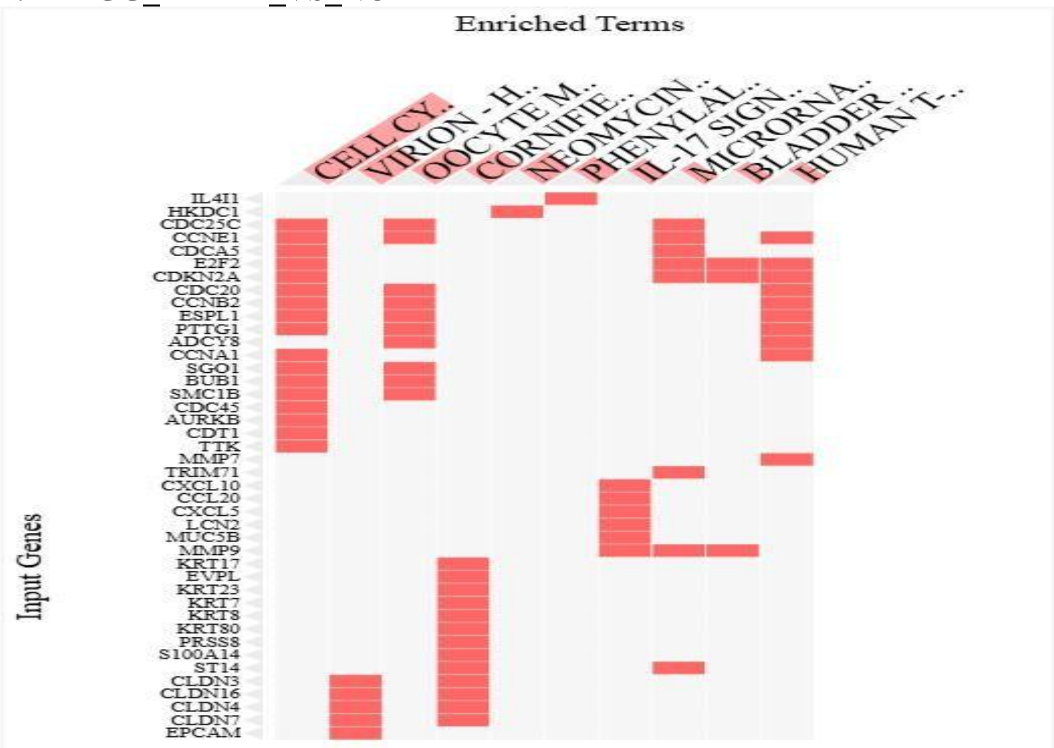


Fig-13: KEGG Pathway Clustergram (Early vs. Normal)

Clustergram representing the top enriched KEGG pathways and their associated differentially expressed genes (DEGs) in the early stage compared to normal tissue. The y-axis lists the Input Genes, while the x-axis displays the Enriched KEGG Terms. Red squares indicate a positive association (presence) of a gene within a specific biological pathway. The hierarchical clustering of terms (top) and genes (left) reveals functional modules, particularly highlighting a dense cluster of cell cycle regulators and structural components of the cornified envelope and epithelial barrier.

1. The **Early vs. Normal** stage KEGG analysis reveals a tissue landscape undergoing a radical shift from homeostatic stability to aggressive, uncontrolled expansion. At the core of this transition is the massive upregulation of the Cell Cycle and DNA Replication pathways. Genes such as *CDC20*, *CCNB2*, and *BUB1* serve as the primary molecular engines, driving the cell through mitotic checkpoints and forcing a hyper-proliferative state. This internal growth surge is complemented by the hijacking of Virion-related signaling pathways, where proteins like *EPCAM* and the *CLDN* (Cloudin) family-specifically *CLDN3* and *CLDN4*-are repurposed to bypass cellular communication limits and facilitate nuclear transport, mirroring mechanisms typically seen in viral infections.

Simultaneously, the tissue undergoes structural and environmental remodeling. The enrichment of the Cornified Envelope and Keratin family genes (*KRT7, KRT17*) indicates a defensive stress response and a loss of epithelial identity. This physical breakdown is further accelerated by the IL-17 signaling pathway, where pro-inflammatory chemokines like *CXCL10* and *MMP9* create a chronic inflammatory microenvironment.

Biological Process

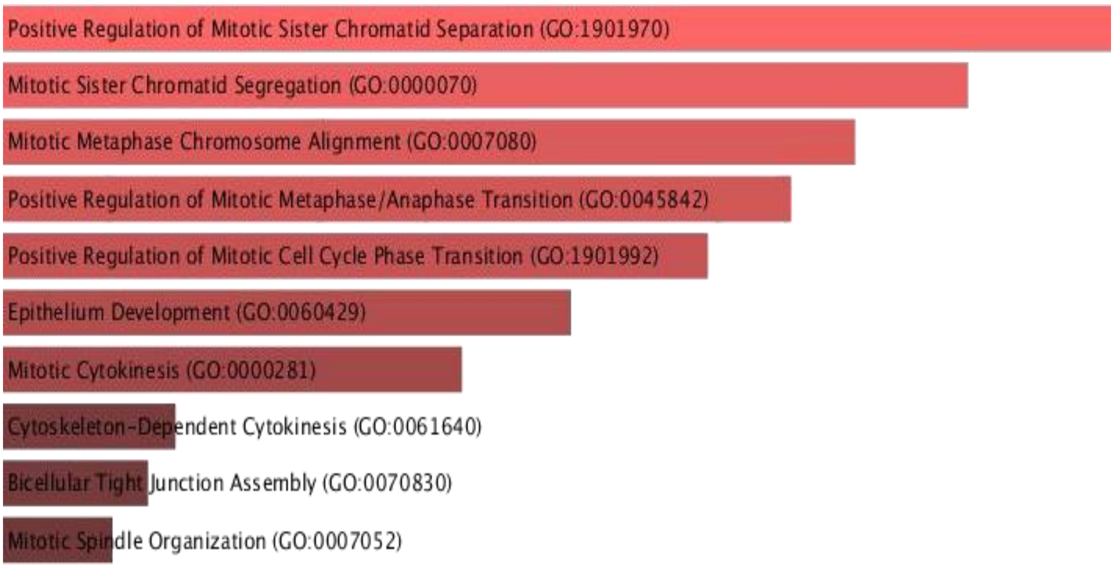


Fig-13(a): **The bar chart illustrates the top enriched biological processes identified through Gene Ontology analysis**. The results are ranked by statistical significance, where bar length correlates with the combined enrichment score and color intensity represents the p-value significance. The profile is dominated by terms related to the mechanical and regulatory phases of the mitotic cell cycle.

2. The Biological Process analysis reveals that the transition from normal to early-stage tissue is primarily driven by an aggressive acceleration of the cell division machinery. The most significant enrichment is observed in the regulation of sister chromatid separation and chromosome alignment, indicating that the

tissue has optimized its ability to replicate genetic material rapidly. Furthermore, the data highlights a shift in the metaphase-to-anaphase transition, suggesting that early-stage cells effectively bypass standard regulatory checkpoints to ensure continuous proliferation. Beyond simple division, there is a notable involvement of epithelium development and cytokinesis pathways. This indicates that the early progression of the disease involves not only an increase in cell number but also a fundamental reorganization of the epithelial layer. The coordination between spindle organization and cytoskeletal-dependent cytokinesis suggests a high level of structural plasticity, allowing the expanding cell population to physically reshape the nascent lesion environment.

Cellular Process

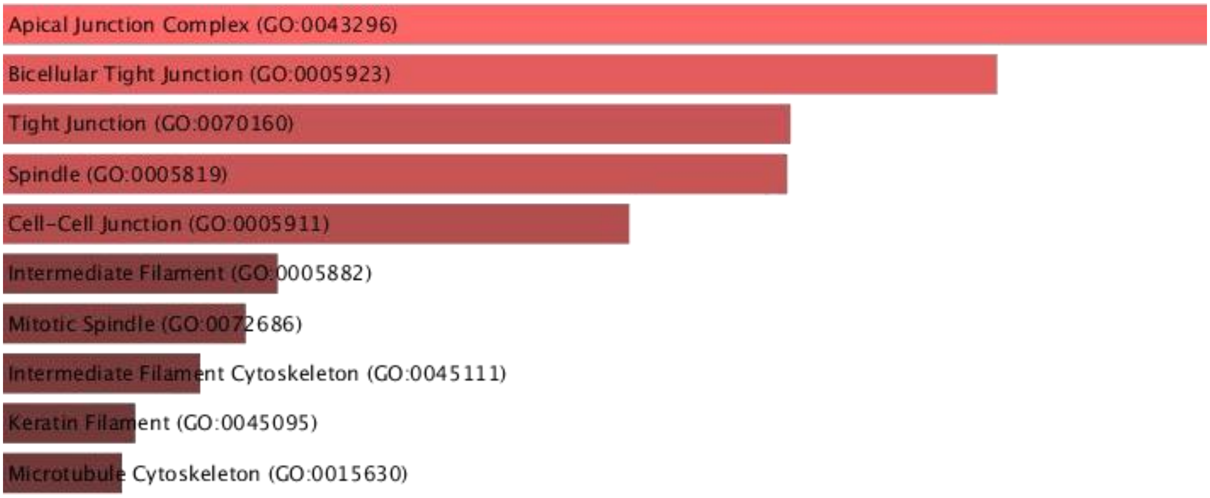


Fig-13(b) This chart displays the cellular locations where the differentially expressed genes are most active. The ranking reflects the compartments of the cell undergoing the most significant proteomic and structural changes during early-stage development, with a specific focus on the cell surface and the mitotic apparatus.

3. Analysis of Cellular Components pinpointed two distinct hubs of activity: the cell-to-cell junctions and the mitotic spindle apparatus. The highest level of enrichment was found within the apical junction complex and bicellular tight junctions. In healthy tissue, these structures maintain epithelial integrity and polarity; however, their prominence in the early-stage results suggests a widespread remodeling of the physical "glue" that holds cells together. This alteration is a critical precursor to the loss of tissue architecture. Simultaneously, significant activity was identified in the spindle and microtubule cytoskeleton. These components are essential for the physical separation of chromosomes, confirming that the cellular architecture is being heavily recruited to support the high mitotic rate identified in the biological process

analysis. Additionally, the involvement of intermediate filaments and keratin filaments points to a shift in the mechanical properties of the cell, potentially facilitating the survival of cells under the physical stress of rapid tissue expansion.

Molecular Process

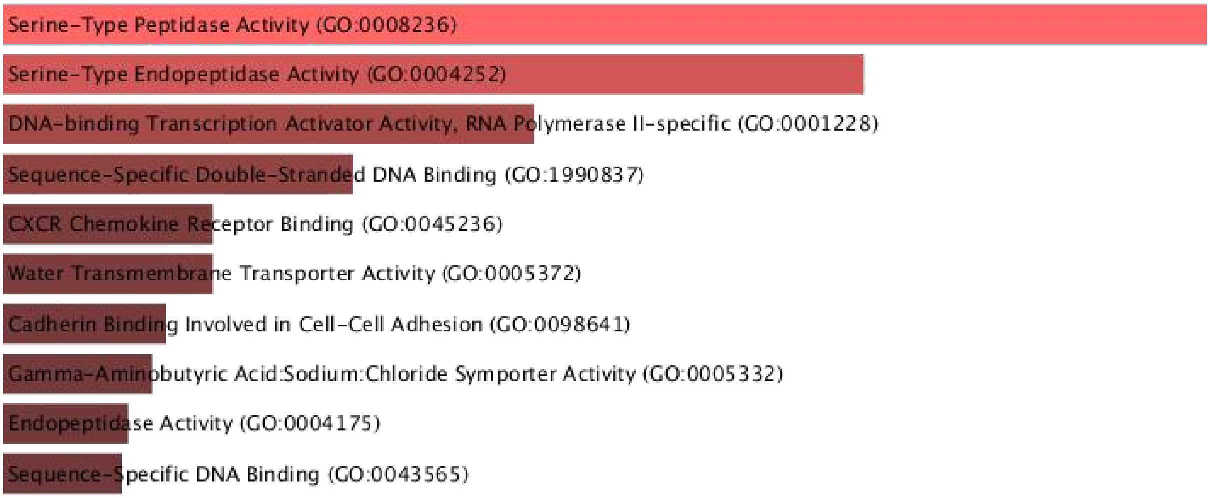


Fig-13(©) The Molecular Function bar chart identifies the specific biochemical activities and binding properties of the upregulated gene products. The results categorize the "molecular tools" being utilized by the cell to drive early-stage progression, ranked by their statistical significance and combined scores.

The Molecular Function profile is characterized by a surge in enzymatic activity and DNA-binding capabilities. The most significant finding is the high enrichment of serine-type peptidase and endopeptidase activities. These enzymes function as molecular "scissors," capable of degrading the extracellular matrix and activating growth factors, which facilitates both tissue expansion and the softening of cellular boundaries. Complementing this enzymatic shift is a robust increase in DNA-binding transcription activator activity and sequence-specific DNA binding. This reflects an intensive regulatory effort within the nucleus to sustain the proliferative transcriptome. Furthermore, the analysis identified significant activity in chemokine receptor binding and cadherin binding. The presence of chemokine signaling suggests that early-stage cells are already developing the capacity to interact with and potentially manipulate the local immune microenvironment. Meanwhile, changes in cadherin binding reinforce the evidence of junctional remodeling, suggesting that the molecular basis for altered cell-to-cell adhesion is firmly established in the early phase of the disease.

late vs normal

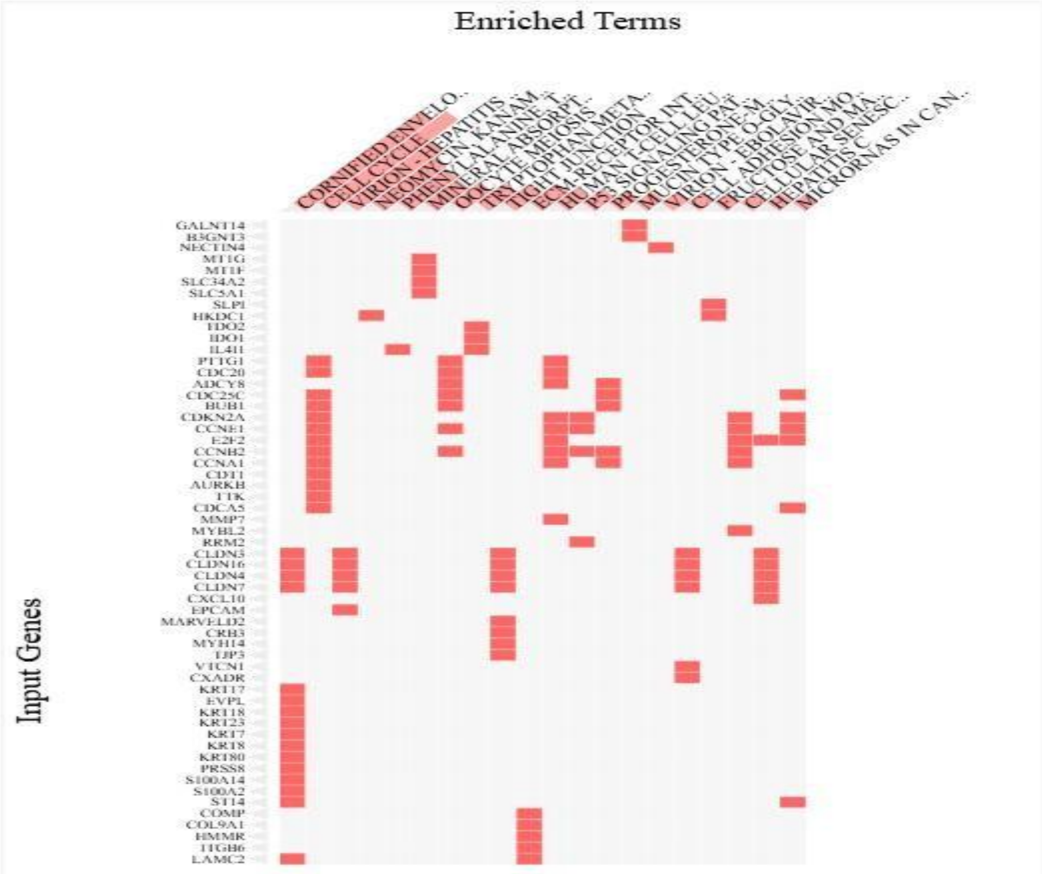


Fig 14: KEGG Pathway Clustergram (Late vs. Normal)

Clustergram illustrating the top enriched KEGG pathways and their corresponding gene associations for the late stage compared to normal control samples. The x- axis represents functionally enriched terms, while the y- axis displays the input gene list. Red indicators signify the presence of specific genes within a pathway. The visualization reveals significant gene-pathway overlaps, particularly highlighting a transitional shift from purely proliferative cycles to complex environmental remodeling, metabolic reprogramming, and advanced signaling mechanisms.

Biological Process

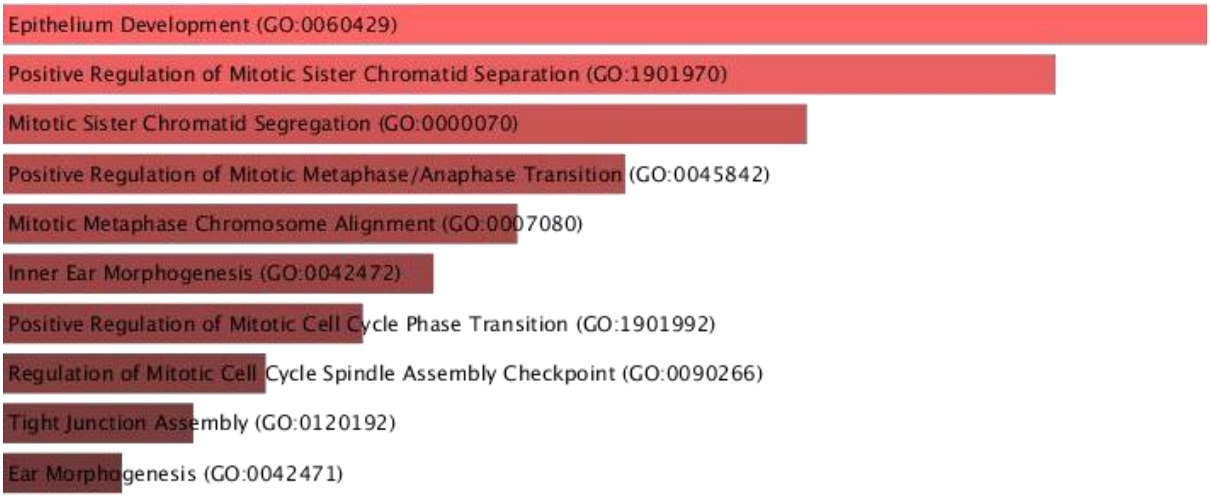


Fig- 14(a) The bar chart represents the top enriched biological processes for late-stage samples compared to normal controls. The results are ranked by their statistical significance, with the horizontal axis reflecting the combined enrichment score. The visualization highlights a significant transition from early-stage cell cycle dominance to more complex processes involving epithelium development and specialized structural assemblies.

The Biological Process analysis for the late stage illustrates a multi-faceted approach to tissue progression. While mitotic processes like sister chromatid separation and metaphase alignment remain significant, the top-ranked term is now Epithelium Development. This suggests that in the late stage, the tissue is undergoing a large-scale architectural overhaul rather than just simple cell doubling. A critical finding is the enrichment of Tight Junction Assembly, which indicates that the cells are actively modifying their permeability and physical connectivity to the surrounding environment. Interestingly, the presence of morphogenesis terms suggests that late-stage cells have reactivated developmental programs that are usually dormant in healthy adult tissue. This "developmental hijacking" allows the tissue to become more plastic and adaptable, facilitating its survival in the hostile environments typical of advanced disease states.

Cellular Process

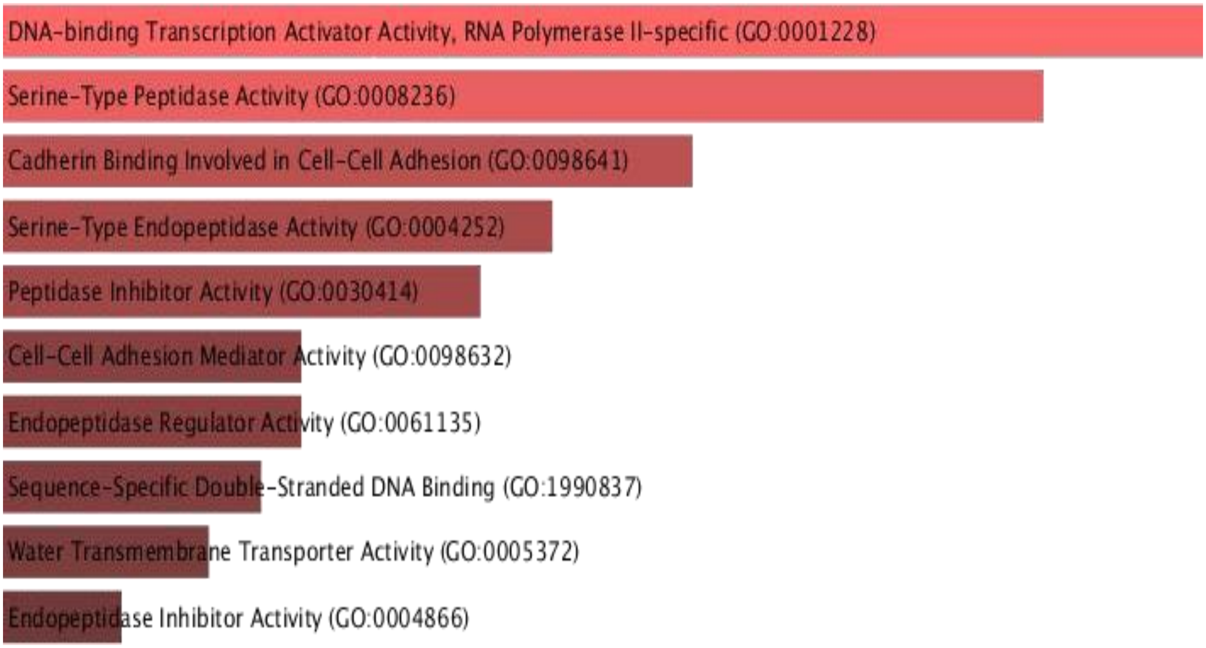


Fig-14(b): This chart identifies the specific cellular compartments and structures where the differentially expressed genes are most active in the late stage. The results focus on the physical boundaries of the cell and the internal scaffolding required for structural integrity and mobility.

The Cellular Component profile for the late stage confirms a profound focus on the cell surface and the cytoskeleton. The Apical Junction Complex and Tight Junctions remain the most enriched compartments, but they are now joined by a strong signal for Intermediate Filaments and the Keratin Filament cytoskeleton. This shift toward keratin-based structural changes suggests that late-stage cells are significantly altering their mechanical stiffness and internal "rigging." Furthermore, the continued enrichment of the Spindle Microtubule and Spindle components indicates that high-rate cell division is still a core feature, even as the cells prioritize structural remodeling. The broad involvement of cell-cell junction terms underscores a comprehensive failure of normal tissue adhesion, likely contributing to the disorganized and invasive growth patterns observed in late-stage progression.

Molecular Process

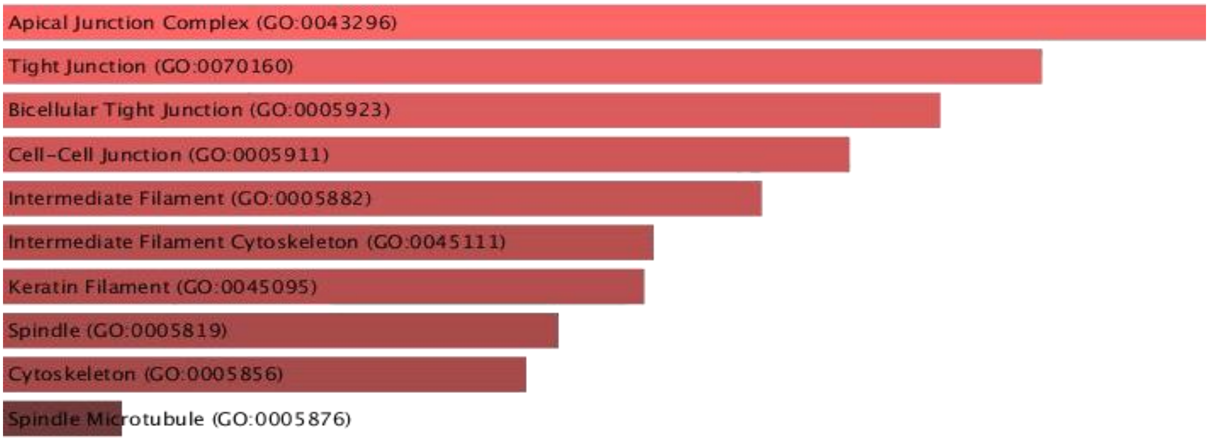


Fig-14(©) The bar chart displays the biochemical functions and binding activities enriched in the late stage. The ranking identifies the molecular "machinery" utilized for environmental degradation, genetic regulation, and cellular communication.

The Molecular Function analysis highlights a sophisticated biochemical arsenal in the late stage. DNA-binding transcription activator activity is the most significant function, indicating a high-level mastery over the cell's genetic expression programs. This is closely followed by Serine-type peptidase activity, which provides the enzymes necessary for degrading the extracellular matrix and clearing physical paths through neighboring tissue. A unique feature of the late-stage molecular profile is the enrichment of Cadherin Binding and Cell-Cell Adhesion Mediator Activity. These functions indicate that the cells are not just breaking junctions, but are actively "re-tooling" how they stick to one another and to new surfaces. Additionally, the presence of Peptidase Inhibitor Activity suggests that late-stage cells have developed protective mechanisms to prevent their own enzymes from causing self-damage, a sign of advanced survival adaptation and metabolic resilience.

Early vs late

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	VIRION - HUMAN IMMUNODEFICIENCY VIRUS	0.005737	0.1101	226.97	1171.32
2	VIRION - FLAVIVIRUS AND ALPHAVIRUS	0.006881	0.1101	181.56	904.00
3	VIRION - EBOLAVIRUS, LYSSAVIRUS AND MORBILLIVIRUS	0.01372	0.1463	82.50	353.87
4	VIRION - LASSA VIRUS AND SFTS VIRUS	0.01938	0.1550	56.71	223.63
5	AMINO SUGAR AND NUCLEOTIDE SUGAR METABOLISM	0.04172	0.2182	25.18	79.99
6	HEDGEHOG SIGNALING PATHWAY	0.06249	0.2182	16.46	45.65
7	GLUTATHIONE METABOLISM	0.06573	0.2182	15.61	42.50
8	DRUG METABOLISM - CYTOCHROME P450	0.07645	0.2182	13.31	34.22
9	CHEMICAL CARCINOGENESIS - DNA ADDUCTS	0.07645	0.2182	13.31	34.22
10	METABOLISM OF XENOBIOTICS BY CYTOCHROME P450	0.08389	0.2182	12.06	29.89

Table-5 early vs late Kegg view in table

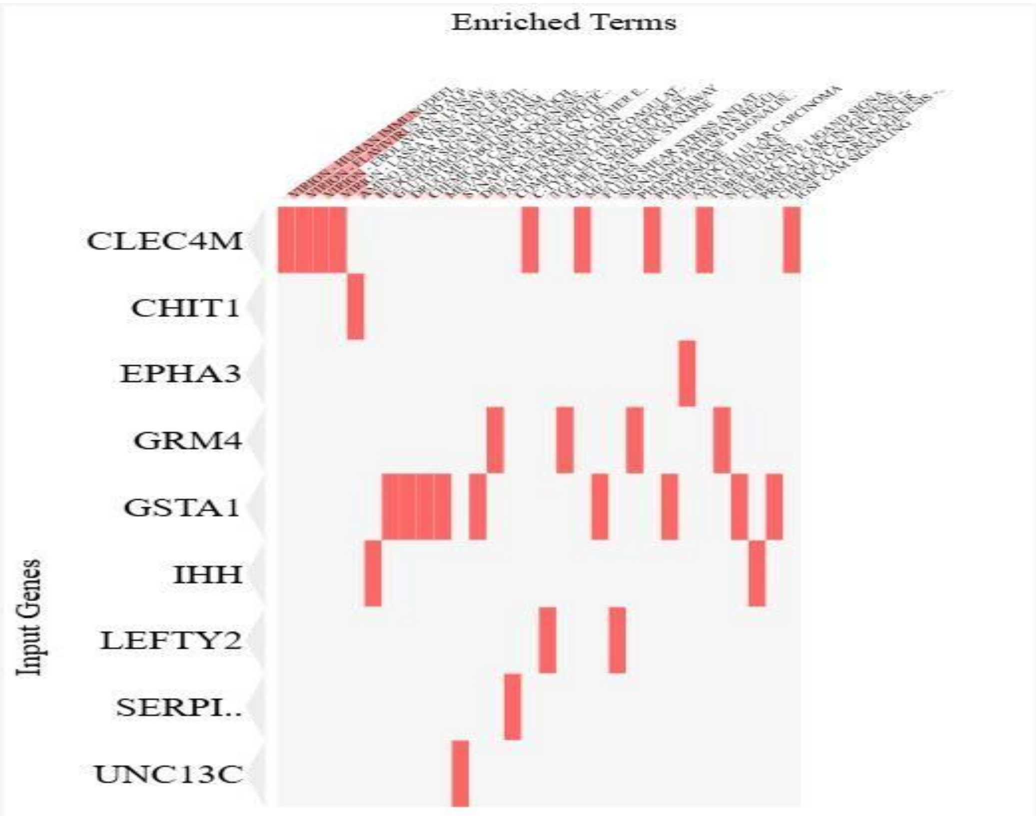


Fig 15: Clustergram and enrichment table of the top 10 KEGG pathways for upregulated genes in the transition from early to late stage. The analysis identifies a specialized transcriptomic signature with a high concentration of genes in viral-mimicry signaling and advanced metabolic pathways. The horizontal axis lists terms ranked by their combined score, while the vertical axis displays the 23-gene input list. The high Odds Ratios (exceeding 200 for top terms) indicate a highly specific enrichment for cellular "hijacking" and detoxification mechanisms during the late-stage transition.

The direct comparison between early and late stages reveals a specialized transcriptomic shift away from simple cell proliferation toward advanced survival and "stem-like" characteristics. The most significant enrichment occurs in pathways categorized as Virion - Human Immunodeficiency Virus and

Flavivirus/Alphavirus. Within this module, the gene CLEC4M acts as a central hub. In this biological context, these findings suggest that late-stage cells have hijacked innate immune signaling and nuclear transport mechanisms to bypass host defenses and facilitate more aggressive growth.

Furthermore, the transition is marked by the activation of the Hedgehog Signaling Pathway, driven by IHH (Indian Hedgehog). This pathway is a master regulator of tissue development and stemness; its reactivation in the late stage suggests that the cells have acquired a more primitive, regenerative, and potentially drug-resistant phenotype. This is complemented by a surge in Glutathione Metabolism and Cytochrome P450 Drug Metabolism, primarily involving the gene GSTA1. These metabolic systems function as an advanced detoxification suite, allowing late-stage cells to survive higher levels of oxidative stress and potentially neutralize therapeutic interventions. Collectively, these 23 genes represent a molecular "escape kit" that

enables the transition from a localized early lesion to a more resilient and metabolically complex late-stage pathology.

GO ONTOLOGY

BIOLOGICAL PROCESS

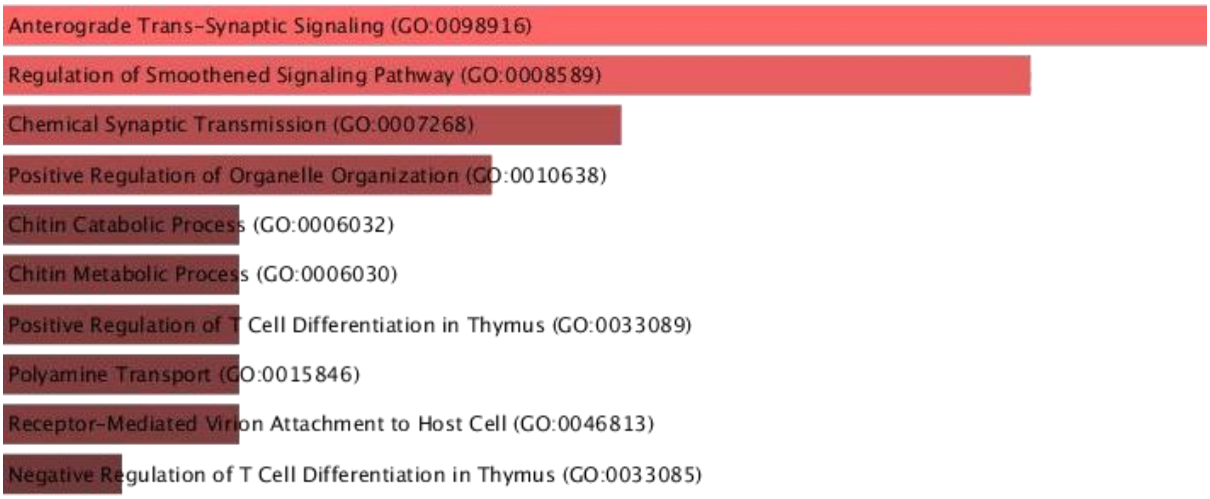


Fig-15(a): The bar chart illustrates the top enriched biological processes upregulated during the transition from the early to the late stage. The results are ranked by statistical significance, with bar length representing the combined score. The functional profile shifts toward advanced signaling pathways and metabolic processes that facilitate cellular communication and environmental adaptation.

The Biological Process analysis reveals that the transition from early to late stage is characterized by a shift toward specialized signaling and metabolic hijacking. The most significant enrichment is observed in Anterograde Trans-Synaptic Signaling and Chemical Synaptic Transmission. In a non-neuronal research context, this indicates that late-stage cells have repurposed high-speed communication machinery to coordinate activities across the tissue mass. A critical driver identified is the Regulation of Smoothened Signaling Pathway, which is a core component of Hedgehog signaling. This pathway is essential for maintaining stem-cell-like properties and tissue remodeling. Furthermore, the enrichment of Chitin Metabolic Processes and Polyamine Transport suggests that late-stage cells have undergone significant metabolic rewiring, likely to support the production of protective extracellular barriers and to manage the high polyamine requirements associated with advanced cellular growth and survival

Cellular Process

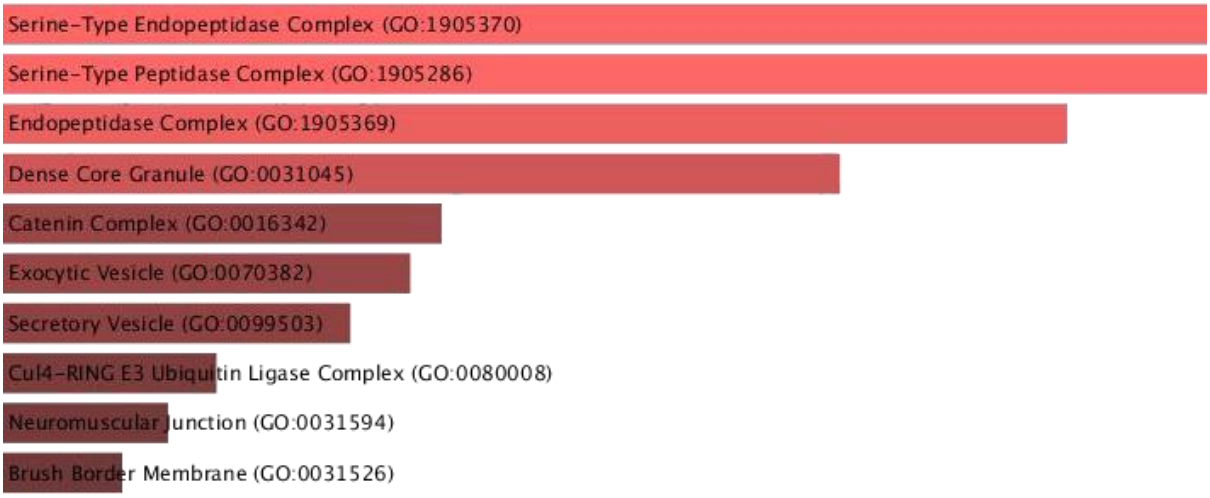


Fig-15(b) Chart displays the cellular locations and complexes where the transition-driving genes are most active. The ranking highlights the compartments involved in protein degradation, secretion, and specialized structural junctions.

Analysis of Cellular Components identifies the Serine-Type Endopeptidase Complex as the primary hub of activity during the early-to-late transition. This concentration of protein-cleaving machinery within specific cellular compartments suggests a highly organized effort to degrade the surrounding tissue matrix. The profile also highlights a significant focus on secretory and transport vesicles, specifically Dense Core Granules, Exocytic Vesicles, and Secretory Vesicles. This indicates that late-stage cells have optimized their "export" machinery to release enzymes and signaling molecules into the microenvironment. Additionally, the involvement of the Catenin Complex and Cul4-RING E3 Ubiquitin Ligase Complex suggests that the transition involves a sophisticated internal "quality control" shift, where specific proteins are targeted for degradation to facilitate the rapid structural and functional changes required for late-stage progression.

Molecular Process

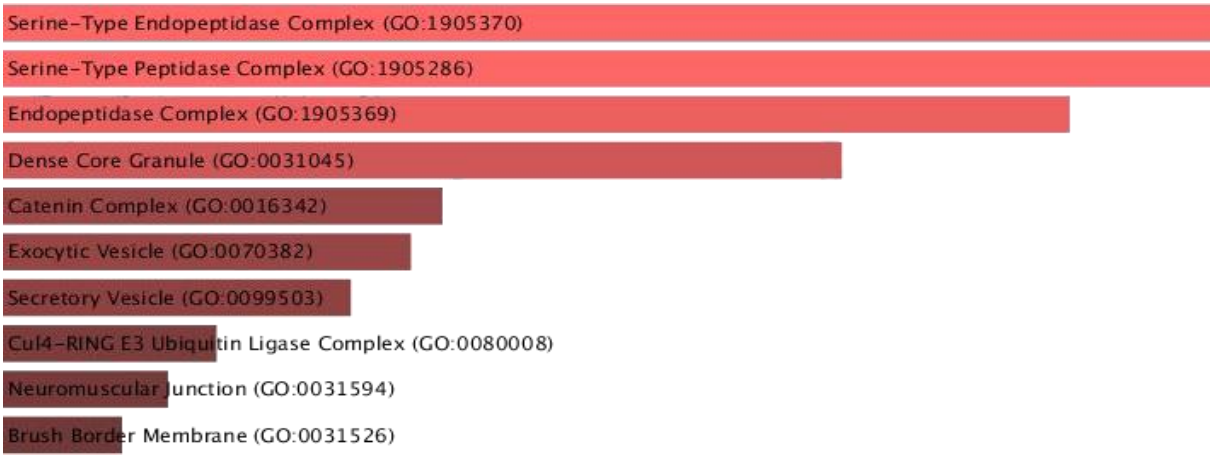


Fig-15 (©) The bar chart identifies the biochemical functions and binding activities that define the transition from early to late stages. The results emphasize the enzymatic tools used for protein processing and the specialized transporters required for advanced metabolic needs.

The Molecular Function profile is dominated by specialized enzymatic activities that facilitate tissue invasion and immune evasion. Mirroring the cellular component data, the top functions are Serine-Type Endopeptidase and Peptidase Activity, providing the chemical "scissors" necessary for advanced environmental remodeling. A notable finding in this transition is the involvement of Receptor-Mediated Virion Attachment to Host Cell. Biologically, this suggests that late-stage cells are utilizing specific surface receptors to "tether" themselves to new environments or to hijack host signaling pathways in a manner similar to viral entry. The analysis also reveals enrichment in Polyamine Transporter Activity and Chitinase Activity, reflecting the metabolic shift toward specialized survival strategies. Collectively, these molecular functions provide the late-stage cells with the necessary tools to navigate, modify, and survive within an increasingly complex and hostile tissue landscape.

The three driver genes not involved I any pathways:

Gene	Early vs. Normal	Late vs. Normal	Early vs. Late (Transition)
TMEM132B	Upregulated	Downregulated	Upregulated
ITLN2	Upregulated	Downregulated	Upregulated
RERGL	Downregulated	Downregulated	Upregulated

Table-6 Three independent gene

Interpretation:

The identification of RERGL, TMEM132B, and ITLN2 as highly differentially expressed genes highlights a unique "molecular switch" mechanism that defines the transition from early to late-stage progression. While these genes are under-represented in current ovarian cancer literature, their dynamic expression profiles suggest they act as stage-specific regulators. RERGL, a member of the Ras GTPase superfamily, shows a progressive decline from early to late stages; its role as a tumor suppressor in other malignancies implies that its loss is a critical requirement for advancing toward a more aggressive phenotype.

In contrast, TMEM132B and ITLN2 exhibit a transient "up-then-down" pattern. TMEM132B, a transmembrane protein involved in structural adhesion, appears to be recruited during the initial tissue reorganization of the early stage but is suppressed during the transition to the late stage, potentially facilitating increased cellular mobility. Similarly, ITLN2 (Intelectin-2) is involved in glycan recognition and immune modulation; its early upregulation followed by late-stage downregulation points toward a sophisticated immune-evasion strategy where the tissue eventually "silences" recognition markers to bypass surveillance. Collectively, the high logFC of these three genes—despite their absence from standard KEGG pathways—identifies them as novel, high-priority candidates for a stage-discriminatory biomarker panel, representing the under-explored temporal biology of disease evolution.

4.5 Downregulated pathway involvement

Early vs Normal Downregulated

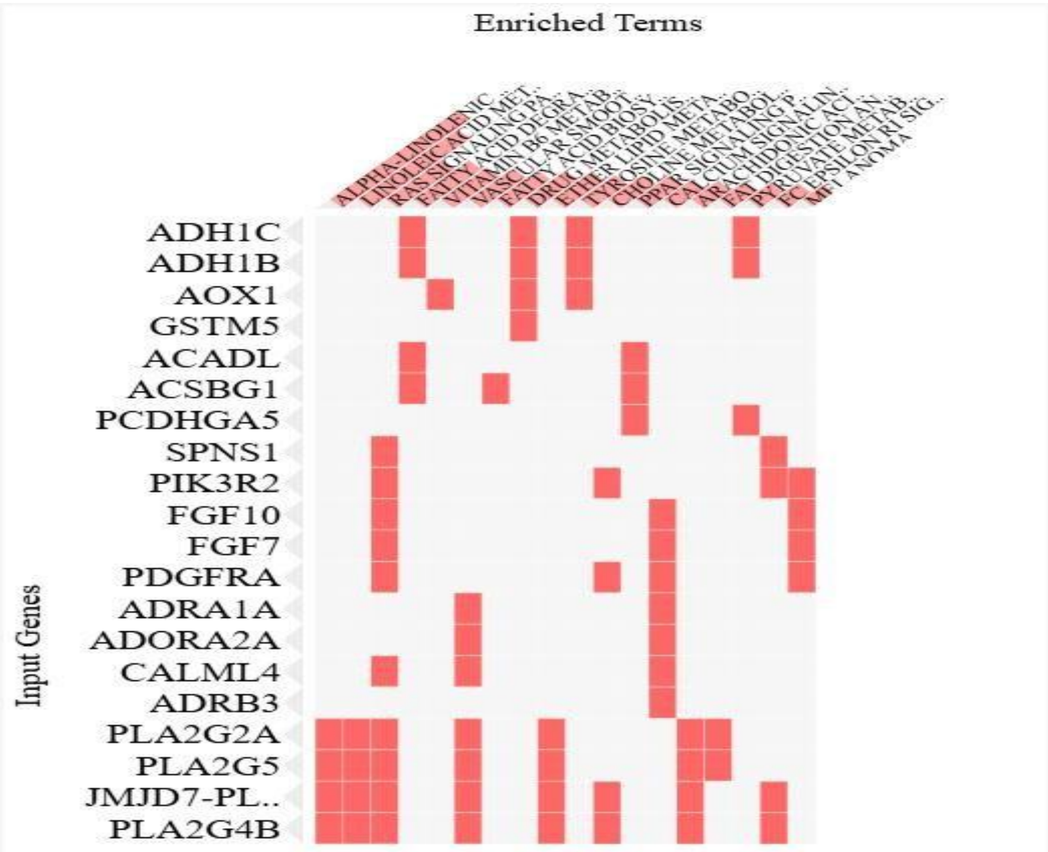


Fig-16 Clustergram of the top enriched KEGG pathways for downregulated genes in the early stage compared to normal control samples. The horizontal axis identifies functionally enriched terms, while the vertical axis lists the downregulated input genes. Red squares indicate a gene's association with a specific metabolic or signaling pathway. The visualization highlights a significant loss of metabolic homeostasis, specifically in lipid processing and detoxification pathways, alongside a decrease in critical growth factor signaling that maintains normal tissue architecture.

The Early vs. Normal downregulated profile represents a significant "metabolic silencing" where the tissue sheds its healthy physiological functions to make room for oncogenic growth. This transition is primarily defined by the suppression of Arachidonic Acid and Alpha-Linolenic Acid Metabolism, where the massive downregulation of the phospholipase family—specifically PLA2G2A, PLA2G5, and PLA2G4B—signals a breakdown in membrane integrity and normal lipid signaling. Simultaneously, the tissue loses its natural detoxification defenses through the downregulation of Drug Metabolism and Tyrosine Metabolism genes, such as ADH1C, ADH1B, and AOX1.

Beyond metabolism, there is a clear withdrawal of homeostatic control mechanisms. The downregulation of PPAR Signaling and Fatty Acid Degradation components like ACADL and ACSBG1 suggests a shift away from efficient energy processing. Crucially, the suppression of specialized growth factors and signaling mediators, including FGF10, FGF7, PDGFRA, and PIK3R2, indicates that the regulatory "brakes" and tissue-identity signals that maintain a normal epithelial state are being systematically deactivated. Collectively, this downregulation creates a permissive environment where the tissue is no longer tethered to its original biological purpose, allowing hyper-proliferative pathways to dominate

Late vs Normal Downregulated pathway

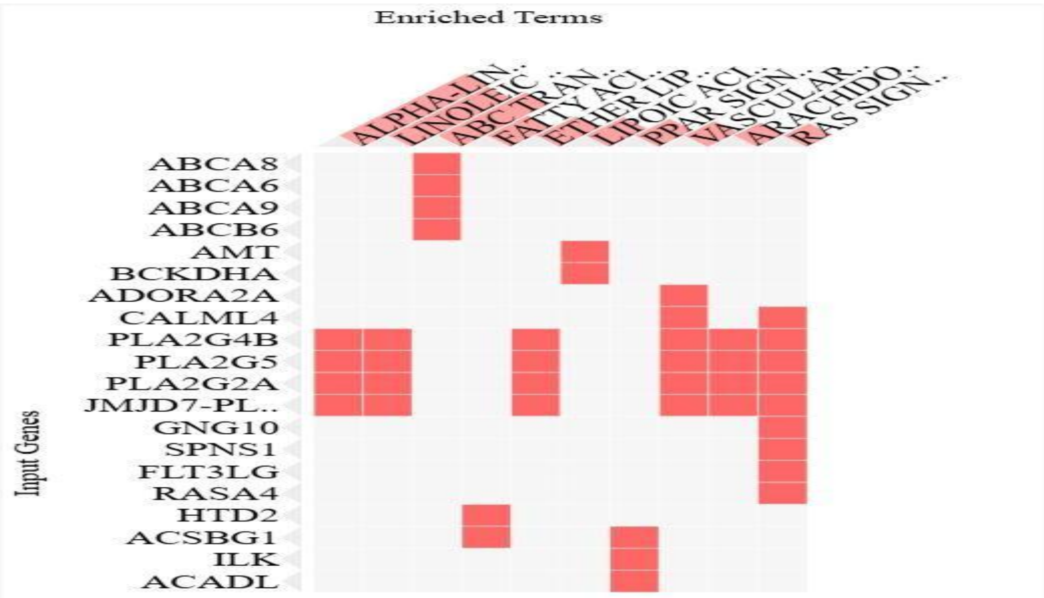


Fig 17 : **Clustergram illustrating the top enriched KEGG pathways for downregulated genes in late-stage samples compared to normal tissue**. The y- axis represents the input gene list, while the x- axis displays enriched functional terms. Red indicators highlight specific gene-pathway memberships. The

results demonstrate a significant loss of transport efficiency, specifically in ABC Transporters, and a collapse of homeostatic lipid and fatty acid signaling networks.

The downregulated profile in the late stage is dominated by the systemic collapse of ABC Transporters, driven by the suppression of genes such as ABCA8, ABCA6, ABCA9, and ABCB6. In healthy tissue, these proteins act as vital "pumps" that move lipids and metabolites across cell membranes; their loss in the late stage suggests a severe disruption in cellular transport and waste management. Furthermore, the persistent downregulation of the phospholipase family -PLA2G4B, PLA2G5, and PLA2G2 -across Alpha-Linolenic, Linoleic, and Arachidonic Acid Metabolism indicates that the loss of anti-inflammatory and membrane-stabilizing lipid signals is a permanent feature of advanced progression.

Additionally, the late-stage tissue shows a critical reduction in energy efficiency and metabolic control. The suppression of Fatty Acid Degradation and PPAR Signaling, involving genes like ACADL, ACSBG1, and HTD2, suggests that the cells have completely abandoned normal oxidative metabolism. The

downregulation of signaling regulators like RASA4 and GNG10 within the Ras Signaling and Vascular Smooth Muscle pathways further implies that the tissue is no longer responsive to the physiological cues that maintain specialized epithelial function. Collectively, these findings suggest that late-stage pathology is characterized by a "locked-in" metabolic state, where the tissue has lost the fundamental transport and signaling tools required for normal biological activity

Early vs late downregulated pathway

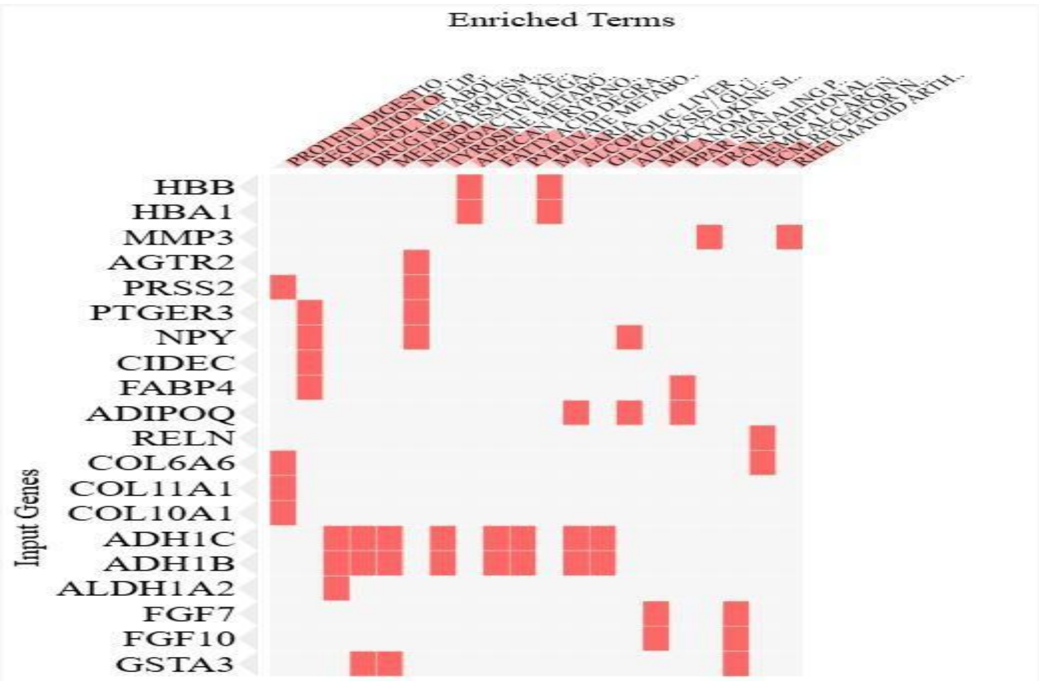


Fig-18: KEGG Pathway Clustergram (Early vs. Late Downregulated)

Clustergram representing the top enriched KEGG pathways for downregulated genes during the transition from the early to late stage. The y-axis lists the downregulated input genes, while the x-axis displays functionally enriched KEGG terms. Red indicators denote the presence of genes within specific pathways. The visualization highlights a significant suppression of metabolic detoxification centers, lipid regulation, and structural collagen pathways, suggesting a systemic decline in tissue-specific homeostatic functions during late-stage progression.

The transition from early to late-stage disease is characterized by a profound suppression of "protective" metabolic and structural pathways. The most significant signature is the collapse of the tissue's detoxification and antioxidant machinery, driven by the dramatic downregulation of the Alcohol Dehydrogenase family (ADH1B, ADH1C) and ALDH1A2. These genes are central to Retinol Metabolism, Drug Metabolism, and the Metabolism of Xenobiotics. Their suppression suggests that as the disease

reaches its late stage, the cells lose their ability to process metabolic byproducts and environmental toxins, likely increasing genomic instability.

Furthermore, the analysis reveals a downregulation of structural and organizational pathways, specifically Protein Digestion and Absorption and ECM-Receptor Interaction, involving genes such as COL11A1, COL10A1, and COL6A6. In healthy or early-stage tissue, these collagens provide structural scaffolding; their loss in the late stage indicates a significant breakdown in tissue architecture. Additionally, there is a clear suppression of the PPAR Signaling Pathway and Regulation of Lipolysis in Adipocytes, involving FABP4, ADIPOQ, and CIDEC. The downregulation of these lipid-handling genes suggests a metabolic "switch" where the late-stage cells abandon normal energy regulation in favor of more specialized, survival-oriented pathways. Finally, the suppression of growth factors like FGF7 and FGF10 indicates that the paracrine signaling that maintains normal epithelial identity has been largely deactivated by the late stage.

4.6 WGCNA Module expesssion anaysis

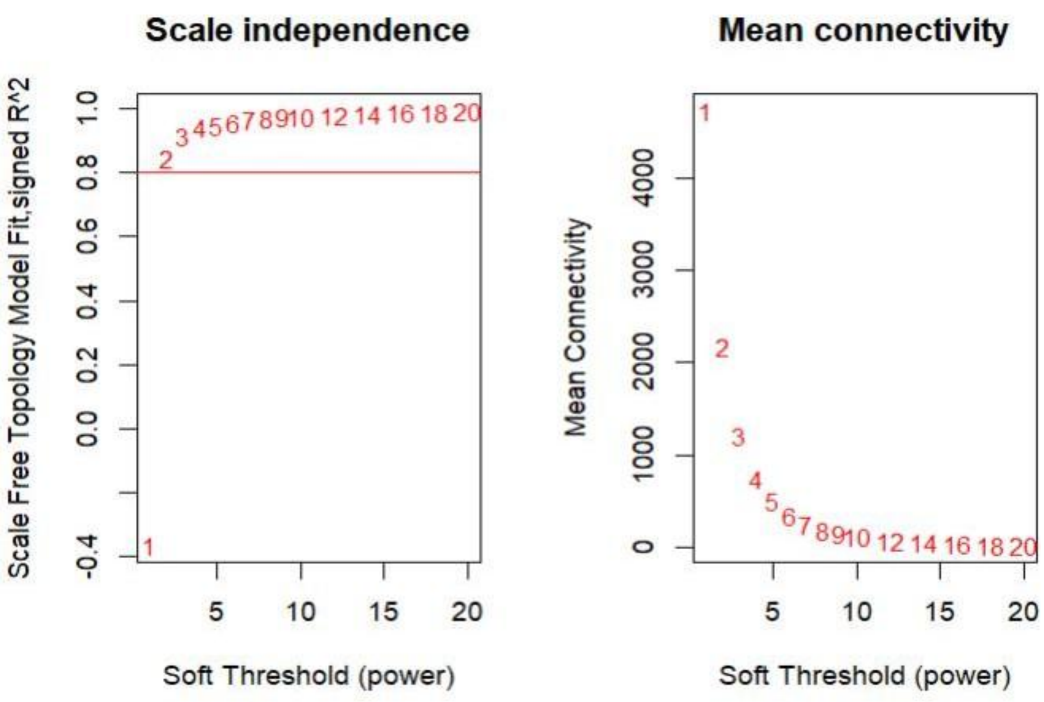


Fig 19: Analysis of Scale-Free Topology Fit Index and Mean Connectivity for Selection of Optimal Soft-Thresholding Power in WGCNA

Analysis of the scale-free fit index (signed R^2) for various soft-thresholding powers beta. The red line indicates the threshold of 0.80. A power of 6 was selected as it reached a scale-free topology fit index of approximately 0.958. (Right) Analysis of the mean connectivity for various soft-thresholding powers. The chosen power of 6 ensures a balance between scale-free topology and sufficient mean connectivity for module detection.

″To construct a biologically meaningful co-expression network, we first evaluated the scale-free topology fit. As shown in **Figure 18**. Consequently, a power of beta = 6 was utilized to produce a network that approximates a scale-free structure while maintaining an acceptable mean connectivity "

Subheading: Identification of Stage-Specific Modules and Hub Genes

Total 22 Modules found

```
> cor_EN[early_modules, ]
       MEblack    MEturquoise          MEpink  MElightgreen        MEgreen
    -0.8125949     -0.9490971      -0.5414455    -0.5369840     -0.6399955
 MElightyellow       MEgrey60        MEyellow   MElightcyan         MEblue
     0.5389943      0.7343201       0.9316993     0.6198096      0.7671999
MEmidnightblue      MEmagenta          MEcyan MEgreenyellow
     0.7198265      0.7636880       0.8876771     0.7640083
```

Fig-20 The total 22 dysregulated modules

The hierarchical clustering of the TOM-based dissimilarity matrix (1-TOM) identified 21 distinct modules, excluding the **Grey** (unassigned) module. **Early-Stage Drivers:** Correlation analysis revealed a robust transcriptional signature in early-stage samples. The **Yellow** module showed the strongest positive correlation ($r = 0.93$, $p < 0.001$), followed by the **Cyan** module ($r = 0.89$). Conversely, the **Turquoise** module (6,757 genes) showed a near-perfect negative correlation ($r = -0.95$).

1. **Early vs Normal Modular Expression**

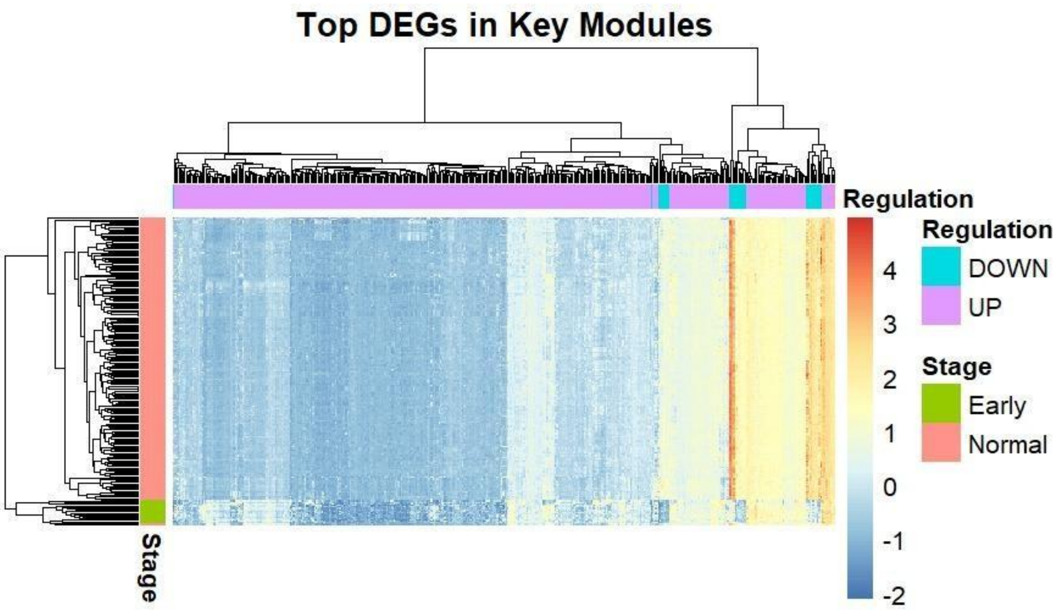


Fig 21 Early vs Normal modules Heatmap

This heatmap illustrates the expression patterns of the top differentially expressed genes (DEGs) identified within key WGCNA modules for the Early vs Normal comparison. Each column represents a gene, and each row corresponds to a sample, with samples annotated by stage (Early and Normal). An additional annotation bar categorizes genes based on their regulation status, distinguishing upregulated (UP) and downregulated (DOWN) genes derived from differential expression analysis.

The color gradient reflects normalized gene expression levels, where red indicates high expression (upregulation), blue indicates low expression (downregulation), and intermediate shades (yellow/white) represent moderate or near-average expression. A clear contrast is observed between Early-stage and Normal samples: genes labeled as upregulated show higher expression (red/yellow) predominantly in Early samples and lower expression (blue) in Normal samples, while downregulated genes display the opposite trend. This opposing pattern highlights strong transcriptional reprogramming occurring during the early stage of disease.

The hierarchical clustering dendrogram demonstrates that samples group according to their biological condition, with Early and Normal samples forming distinct clusters, indicating robust separation based on gene expression profiles. Similarly, genes cluster into groups with similar expression behaviors, suggesting coordinated regulation within modules.

Importantly, the genes included in this heatmap are not arbitrary DEGs but are derived from biologically meaningful co-expression modules identified through WGCNA and further refined by DEG overlap, strengthening their relevance. These genes likely represent key drivers or markers of early-stage disease and may serve as potential candidates for early diagnosis, prognosis, or therapeutic targeting. Overall, this figure provides strong evidence of distinct molecular signatures that characterize early-stage conditions compared to normal tissue.

2. Late vs Normal Modular Heatmap

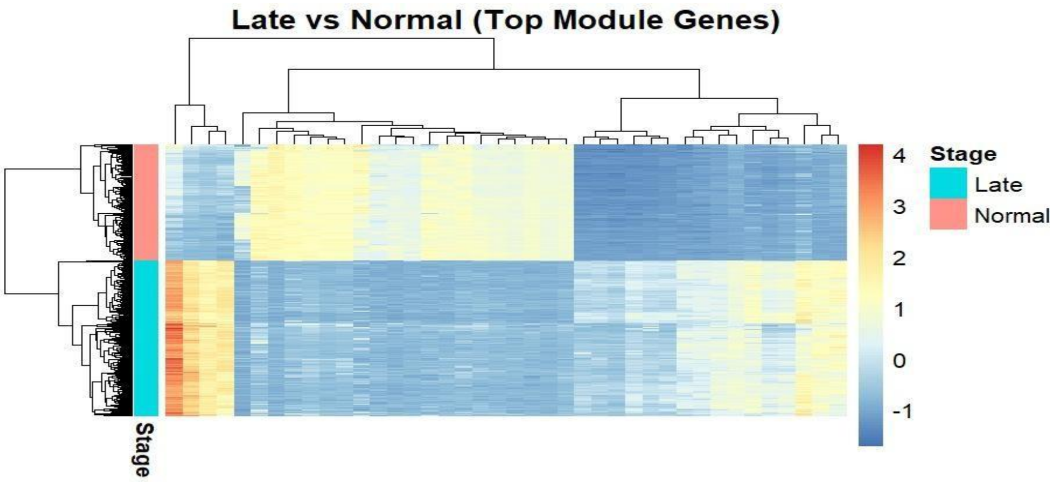


Fig 22:Late vs Normal modular Heatmap

This heatmap represents the expression patterns of selected key genes identified from Late vs Normal comparison, highlighting clear transcriptional differences associated with disease progression. Each column corresponds to a gene, while each row represents a sample, grouped into Late-stage tumor and Normal tissues, as indicated by the stage annotation bar. The color gradient reflects normalized expression levels, where red indicates higher expression (upregulation) and blue indicates lower expression (downregulation).

The genes included in this heatmap are derived from significant WGCNA modules (such as MEblue and MEturquoise) that also overlap with differentially expressed genes, indicating that these genes are not only statistically significant but also biologically relevant. The clear separation and contrasting expression blocks imply that these genes may play critical roles in tumor advancement, potentially serving as stage-specific biomarkers or therapeutic targets."The heatmap uses a color gradient where red indicates upregulation, blue indicates downregulation, and intermediate colors such as yellow represent moderate expression. This pattern highlights distinct gene expression differences between Late-stage and Normal samples

3. Early vs Late no proper modules were found.

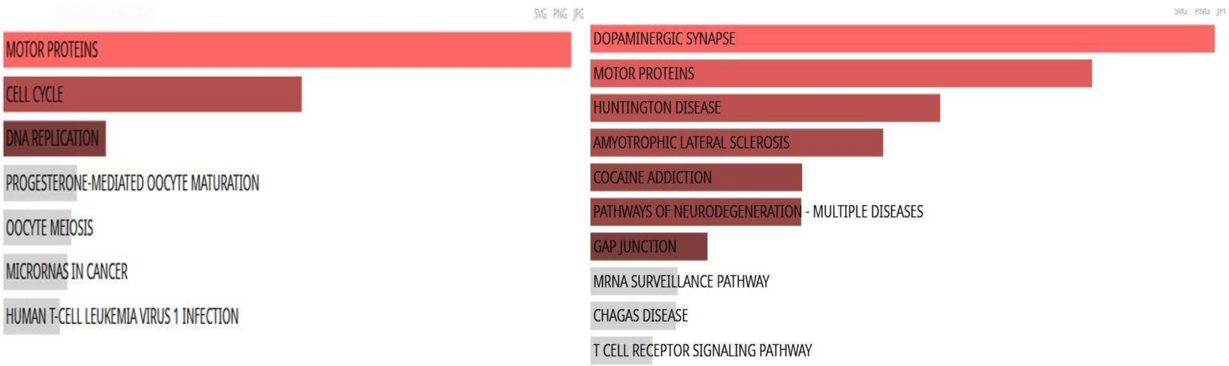


Fig-23 Futher functional enrichment analysis has been done for these top modules of both stage

In the early-stage network, the yellow module, comprising 198 genes overlapping with DEGs, showed that the top 25 genes were predominantly enriched in cell cycle-related pathways, indicating that dysregulation of proliferative processes is a major driving force in early disease development. However, not all genes within this module were confined to a single pathway; a substantial proportion of the remaining genes were associated with diverse biological processes, reflecting the functional heterogeneity of co-expression modules. Similarly, the magenta module, containing 113 genes, exhibited limited but specific enrichment, with only 11 genes mapping to the motor protein pathway, suggesting a more specialized role in cytoskeletal dynamics and intracellular transport rather than broad pathway dominance.

In contrast, late-stage modules demonstrated a shift in biological processes. The turquoise module was predominantly enriched in oxidative phosphorylation, indicating enhanced metabolic activity and energy production associated with tumor progression. characteristic of advanced disease stages.

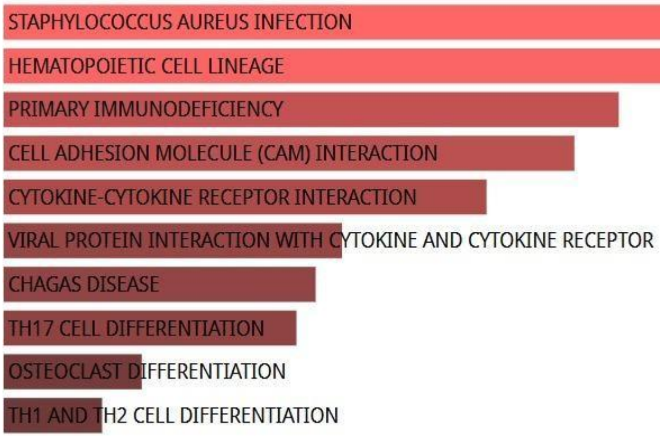


Fig-24 Late Blue Module

Additionally, the late blue module genes were associated with CAM signaling pathways, suggesting alterations in cellular signaling and homeostasis during progression. Overall, these findings highlight a transition from cell cycle-driven mechanisms in early stages to metabolic and signaling adaptations in late stages, underscoring the dynamic nature of disease progression at the molecular level.

CHAPTER-5
CONCLUSION

7.CONCLUSION

This study performed a comprehensive transcriptomic analysis to identify differentially expressed genes associated with ovarian cancer progression by integrating RNA-seq data from TCGA and GTEx. To address technical variability between datasets, batch effect correction using surrogate variable analysis (SVA) was applied, and Principal Component Analysis (PCA) was performed to assess data structure and normalization efficiency.Differential expression analysis using DESeq2 identified a large number of significantly dysregulated genes in both early and late stages of ovarian cancer. The presence of a substantial overlap between early-stage (1931 genes) and late-stage (1999 genes) upregulated genes, with 1739 genes common between them, indicates that core tumor-associated molecular changes are established early and persist throughout disease progression. In contrast, only a small subset of genes (23 genes) was differentially expressed between early and late stages, suggesting that progression is driven by a limited number of critical regulatory genes rather than widespread transcriptional changes.

Among these, specific genes such as **RERGL, EN2, ITLN2, and TMEM132B**, which exhibited high expression changes (log_2FoldChange > 2), were identified in the early versus late-stage comparison. Notably, these genes have been minimally explored in the context of ovarian cancer, highlighting their potential as novel biomarkers for disease progression and early detection.Venn diagram analysis further supported these findings, identifying 183 genes unique to the early stage as potential early diagnostic markers and 255 genes specific to the late stage associated with advanced tumor characteristics. A small subset of progression-associated genes, including a few common across all comparisons, may play a crucial role in the transition from early to late stages.

Integration of differential expression analysis with Weighted Gene Co-expression Network Analysis (WGCNA) revealed stage-specific gene modules and biological processes. Early-stage modules were predominantly associated with cell cycle regulation and proliferative mechanisms, indicating that uncontrolled cell division is a key feature of tumor initiation. In contrast, late-stage modules showed enrichment in oxidative phosphorylation and metabolic pathways, suggesting metabolic reprogramming and increased energy demands during tumor progression. Additionally, pathways related to calcium signaling and immune responses were observed, reflecting changes in cellular signaling and tumor microenvironment interactions.

Functional enrichment analysis using GO and KEGG further confirmed distinct biological patterns between stages, with early-stage tumors showing involvement in cell adhesion and immune-related pathways, while late-stage tumors were enriched in epithelial differentiation and inflammatory processes.Overall, this study demonstrates that ovarian cancer progression involves a transition from proliferation-driven mechanisms in early stages to metabolic and signaling adaptations in advanced stages. The identification of novel candidate genes, particularly **RERGL, EN2, ITLN2, and TMEM132B**, provides new insights into the molecular basis of disease progression. Given the limited existing literature on these genes in ovarian cancer, they represent promising targets for further experimental validation and potential development as diagnostic or prognostic biomarkers.

This integrative approach highlights the importance of combining large-scale transcriptomic data, differential expression analysis, and network-based methods to uncover biologically meaningful insights. The findings of this study lay a strong foundation for future research aimed at improving early detection, understanding disease mechanisms, and developing targeted therapeutic strategies in epithelial ovarian cancer.

CHAPTER 6
FUTURE DIRECTION

6.Future Directions

While this study provides important insights into the molecular mechanisms of ovarian cancer progression, several directions can be pursued to strengthen and extend these findings.

First, the candidate genes identified in this study-**RERGL, EN2, ITLN2, and TMEM132B-**require **further in-depth analysis**, as there are limited studies exploring their roles in ovarian cancer. Future work should focus on validating their expression patterns and investigating their biological functions to determine their involvement in tumor initiation, progression, and metastasis.

Second, **cross-platform validation using independent datasets**, particularly from public repositories such as GEO, should be performed to ensure the reproducibility and robustness of the identified gene signatures. Validating results across different platforms and cohorts will strengthen the reliability of these biomarkers.

Third, **survival analysis** (e.g., Kaplan–Meier analysis) should be conducted to evaluate the prognostic significance of the identified genes. This will help determine whether the expression levels of these genes are associated with overall survival or disease-free survival in ovarian cancer patients.

Fourth, integration of these biomarkers with existing clinical indicators such as CA-125 and HE4 could be explored to develop **improved diagnostic models** with higher sensitivity and specificity for early-stage detection.

Finally, the incorporation of **advanced bioinformatics approaches**, including machine learning models and network-based analyses, can further refine biomarker selection and improve predictive accuracy. Expanding the study to include multi-omics data (proteomics, metabolomics, and epigenomics) may also provide a more comprehensive understanding of ovarian cancer progression.

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